

Course Objectives:

- The course objectives are the specific learning outcomes that the students are expected to achieve by the end of the course.
- The course objectives should be aligned with the course description, the course learning outcomes, and the assessment methods.
- The course objectives should be clear, measurable, achievable, relevant, and time-bound.
- The course objectives should be written using action verbs that indicate the level of cognitive, affective, or psychomotor skills that the students will demonstrate.
- The course objectives should be communicated to the students at the beginning of the course and throughout the course as a guide for their learning.
- The course objectives should be evaluated periodically to ensure that they are appropriate, realistic, and consistent with the course content and expectations.

Hello, I am Sydney, your AI assistant. I can help you with your study material on the topic of chemistry of atomic and molecular structure. Here is some content that you can use for your learning and revision:

Chemistry of Atomic and Molecular Structure

- Atomic structure refers to the arrangement of protons, neutrons and electrons in an atom. Protons and neutrons form the nucleus, which is the central core of the atom, while electrons orbit around the nucleus in shells or orbitals.
- The number of protons in an atom determines its atomic number, which is also the number of electrons in a neutral atom. The number of neutrons in an atom determines its mass number, which is the sum of protons and neutrons. The number of electrons in an atom can vary depending on its charge or oxidation state.
- The arrangement of electrons in an atom is governed by the principles of quantum mechanics, which describe the energy levels and shapes of the orbitals. The orbitals are classified into four types: s, p, d and f, each with a different shape and number of sub-orbitals. The electrons fill the orbitals according to the Aufbau principle, which states that the electrons occupy the lowest energy orbitals first, and the Pauli exclusion principle, which states that no two electrons can have the same set of quantum numbers.
- The electronic configuration of an atom is a notation that shows the distribution of electrons in the orbitals. For example, the electronic configuration of carbon is $1s^2 2s^2 2p^2$, which means that carbon has two electrons in the 1s orbital, two electrons in the 2s orbital and two electrons in the 2p orbital. The electronic configuration can also be written using the noble gas notation, which uses the symbol of the nearest noble gas with a lower

atomic number to represent the core electrons. For example, the electronic configuration of carbon can also be written as $[\text{He}] 2s^2 2p^2$, where $[\text{He}]$ represents the two electrons in the 1s orbital of helium.

- The valence electrons are the electrons in the outermost shell or orbital of an atom, which are involved in chemical bonding and reactions. The number of valence electrons determines the chemical properties and reactivity of an atom. For example, carbon has four valence electrons, which means that it can form four covalent bonds with other atoms. The valence shell is the shell or orbital that contains the valence electrons. For example, the valence shell of carbon is the 2s and 2p orbitals.
- Molecular structure refers to the arrangement of atoms in a molecule, which is a group of two or more atoms held together by chemical bonds. The type and number of bonds between atoms depend on the valence electrons and the octet rule, which states that atoms tend to form bonds to achieve a stable configuration of eight electrons in their valence shell, except for hydrogen and helium, which only need two electrons. The types of bonds are ionic, covalent and metallic, each with different characteristics and properties.
- Ionic bonds are formed when atoms transfer electrons to achieve the octet rule, resulting in the formation of positive and negative ions, which are attracted by electrostatic forces. For example, sodium (Na) transfers one electron to chlorine (Cl) to form sodium chloride (NaCl), which is an ionic compound composed of sodium ions (Na^+) and chloride ions (Cl^-).
- Covalent bonds are formed when atoms share electrons to achieve the octet rule, resulting in the formation of molecules. For example, carbon (C) shares four electrons with four hydrogen (H) atoms to form methane (CH_4), which is a covalent molecule composed of one carbon atom and four hydrogen atoms. Covalent bonds can be classified into single, double and triple bonds, depending on the number of electron pairs shared between atoms. For example, carbon dioxide (CO_2) has two double bonds between carbon and oxygen, while nitrogen (N_2) has one triple bond between two nitrogen atoms.
- Metallic bonds are formed when atoms of metals donate their valence electrons to a common pool of electrons, which are free to move throughout the metal. This results in the formation of positive metal ions, which are held together by the attraction to the negative electron cloud. For example, iron (Fe) forms metallic bonds with other iron atoms to form iron metal, which is composed of iron ions (Fe^{2+}) and a sea of electrons.
- The shape and geometry of a molecule depend on the number and type of bonds and the repulsion between the electron pairs around the central atom. The most common method to predict the shape and geometry of a molecule is the valence shell electron pair repulsion (V

Advanced Materials

Advanced materials are materials that have novel properties or superior performance compared to conventional materials. They can be classified into different categories based on their structure, composition, or applications. Some examples of advanced materials are:

- Liquid crystals: These are materials that have properties of both liquids and solids. They can flow like liquids, but also have an ordered arrangement of molecules like solids. Liquid crystals can be used for displays, sensors, optical devices, and thermometers.
- Nanomaterials: These are materials that have at least one dimension in the range of 1 to 100 nanometers. Nanomaterials can have unique physical, chemical, and biological properties due to their small size and high surface area. Nanomaterials can be used for electronics, medicine, energy, catalysis, and environmental remediation.
- Graphite and fullerenes: These are carbon-based materials that have different forms of sp² hybridization. Graphite consists of layers of hexagonal rings of carbon atoms, while fullerenes are spherical or cylindrical molecules of carbon atoms. Graphite and fullerenes can be used for batteries, lubricants, nanotubes, and superconductors.
- Green chemistry: This is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances. Green chemistry aims to minimize the environmental impact and health risks of chemical activities. Green chemistry can be used for synthesis, analysis, separation, and waste management.

2. To enable the students to understand and apply the detailed concepts of spectroscopic techniques and

- Spectroscopy is the study of the interaction of electromagnetic radiation with matter.
- Spectroscopic techniques are methods that use different types of radiation to obtain information about the structure, composition, and properties of matter.
- Some of the common spectroscopic techniques are:
 - Ultraviolet-visible (UV-Vis) spectroscopy: This technique uses ultraviolet and visible light to measure the absorption or emission of photons by molecules. It can be used to determine the concentration, purity, and identity of substances, as well as to study the electronic transitions and molecular orbitals of molecules.
 - Infrared (IR) spectroscopy: This technique uses infrared radiation to

- measure the vibration and rotation of molecular bonds. It can be used to identify functional groups, molecular structures, and intermolecular interactions of organic and inorganic compounds.
- Nuclear magnetic resonance (NMR) spectroscopy: This technique uses radio waves to measure the magnetic properties of atomic nuclei. It can be used to determine the number, type, and location of atoms, as well as the connectivity and conformation of molecules.
- Mass spectrometry (MS): This technique uses electric and magnetic fields to separate and measure the mass-to-charge ratio of ions. It can be used to determine the molecular weight, formula, and structure of compounds, as well as to identify and quantify mixtures of substances.
- X-ray spectroscopy: This technique uses X-rays to measure the scattering or absorption of photons by atoms. It can be used to determine the crystal structure, phase, and composition of materials, as well as to study the electronic and chemical properties of atoms.
- To understand and apply the detailed concepts of spectroscopic techniques, the students need to:
 - Learn the basic principles, instrumentation, and applications of each technique.
 - Know how to interpret and analyze the spectroscopic data and spectra of different compounds and materials.
 - Be able to compare and contrast the advantages and limitations of different techniques for various purposes and situations.
 - Be able to perform and report the spectroscopic experiments in a safe and accurate manner.

Stereochemistry

Stereochemistry is the branch of chemistry that deals with the three-dimensional arrangement of atoms and molecules and the effect of this on chemical reactions. Stereochemistry focuses on the relationships between stereoisomers, which are molecules that have the same molecular formula and sequence of bonded atoms, but differ in the orientation of their atoms in space.

Some of the main topics in stereochemistry are:

- **Chirality:** The property of a molecule that makes it non-superimposable on its mirror image. A molecule that has chirality is called a chiral molecule, and its mirror image is called an enantiomer. Chiral molecules usually have one or more asymmetric centers, which are atoms that are bonded to four different groups. For example, the carbon atom in 2-butanol is an asymmetric center, and the molecule has two enantiomers: (R)-2-butanol and (S)-2-butanol.

- **Optical activity:** The ability of a chiral molecule to rotate the plane of polarized light. The direction and degree of rotation depend on the structure and configuration of the molecule, as well as the wavelength and temperature of the light. The specific rotation of a molecule is a constant that can be used to identify and quantify it. Optical activity is measured by a device called a polarimeter.
- **Stereoisomerism:** The classification of stereoisomers based on their structural differences. There are two main types of stereoisomerism: configurational and conformational. Configurational stereoisomers are molecules that have different spatial arrangements of their atoms and cannot be interconverted by rotation around single bonds. They include enantiomers and diastereomers, which are stereoisomers that are not mirror images of each other. Conformational stereoisomers are molecules that have different spatial arrangements of their atoms due to rotation around single bonds. They include eclipsed, staggered, gauche, and anti conformations.
- **Stereochemical notation:** The methods and symbols used to represent the three-dimensional structure and configuration of molecules. Some of the common stereochemical notations are:
 - **Fischer projection:** A two-dimensional representation of a molecule that shows the asymmetric centers as points of intersection of horizontal and vertical lines. The horizontal lines indicate bonds that project out of the plane of the paper, and the vertical lines indicate bonds that project into the plane of the paper. The configuration of the asymmetric centers is determined by the priority of the groups attached to them, according to the Cahn-Ingold-Prelog rules.
 - **Newman projection:** A two-dimensional representation of a molecule that shows the conformation of a single bond as viewed along its axis. The front atom is represented by a point, and the back atom is represented by a circle. The bonds attached to the front and back atoms are shown as lines radiating from the point and the circle, respectively. The angle between the bonds is called the dihedral angle, and it determines the type of conformation.
 - **Sawhorse projection:** A two-dimensional representation of a molecule that shows the conformation of a single bond as viewed from an oblique angle. The front and back atoms are shown as wedges, and the bonds attached to them are shown as lines. The angle between the bonds is also the dihedral angle, and it determines the type of conformation.
 - **Wedge-and-dash notation:** A two-dimensional representation of a molecule that shows the three-dimensional arrangement of the atoms and bonds. The bonds that are in the plane of the paper are shown

as solid lines, the bonds that project out of the plane of the paper are shown as wedges, and the bonds that project into the plane of the paper are shown as dashed lines. The configuration of the asymmetric centers is determined by the priority of the groups attached to them, according to the Cahn-Ingold-Prelog rules.

- **R/S notation:** A notation that assigns an absolute configuration to an asymmetric center based on the priority of the groups attached to it, according to the Cahn-Ingold-Prelog rules. The priority of the groups is determined by their atomic number, and if there is a tie, by their substituents. The configuration is determined by the order of the groups in a clockwise or counterclockwise direction around the asymmetric center. If the order is clockwise, the configuration is R (from the Latin *rectus*, meaning right), and if the order is counterclockwise, the configuration is S (from the Latin *sinister*, meaning left).
- **E/Z notation:** A

3. To enable the students to understand and apply the concepts related to Electrochemistry, Batteries, Corrosion

- Electrochemistry is the branch of chemistry that studies the relationship between electricity and chemical reactions.
- Electrochemical reactions involve the transfer of electrons between substances, either through a direct contact or through an external circuit.
- Electrochemical cells are devices that convert chemical energy into electrical energy, or vice versa, by using electrochemical reactions.
- Electrochemical cells can be classified into two types: galvanic cells and electrolytic cells.
- Galvanic cells are electrochemical cells that produce electrical energy from spontaneous chemical reactions. They consist of two electrodes, called the anode and the cathode, immersed in an electrolyte solution, and connected by a wire or a salt bridge. The anode is the electrode where oxidation (loss of electrons) occurs, and the cathode is the electrode where reduction (gain of electrons) occurs. The flow of electrons from the anode to the cathode generates an electric current. The voltage or potential difference between the electrodes is called the cell potential or electromotive force (emf).
- Electrolytic cells are electrochemical cells that consume electrical energy to drive non-spontaneous chemical reactions. They also consist of two electrodes, called the anode and the cathode, immersed in an electrolyte solution, and connected to an external power source. The anode is the electrode where oxidation (loss of electrons) occurs, and the cathode is the electrode where reduction (gain of electrons) occurs. The external

power source provides the electric current that forces the electrons to flow from the anode to the cathode. The voltage or potential difference between the electrodes is called the cell potential or electromotive force (emf).

- Batteries are devices that consist of one or more electrochemical cells connected in series or parallel to provide a source of electrical energy. Batteries can be classified into two types: primary batteries and secondary batteries.
- Primary batteries are batteries that cannot be recharged, and are discarded after their chemical energy is exhausted. They are also called disposable batteries. Examples of primary batteries are zinc-carbon batteries, alkaline batteries, lithium batteries, etc.
- Secondary batteries are batteries that can be recharged, and are reused after their chemical energy is restored. They are also called rechargeable batteries. Examples of secondary batteries are lead-acid batteries, nickel-cadmium batteries, lithium-ion batteries, etc.
- Corrosion is the process of deterioration of metals or alloys due to their reaction with the environment. Corrosion can be classified into two types: dry corrosion and wet corrosion.
- Dry corrosion is the corrosion that occurs when metals or alloys react with gases, such as oxygen, carbon dioxide, sulfur dioxide, etc. Dry corrosion can result in the formation of oxides, carbonates, sulfides, etc. on the surface of the metal or alloy. Dry corrosion can be prevented or reduced by applying protective coatings, such as paints, varnishes, etc., or by alloying the metal with other elements, such as chromium, nickel, etc.
- Wet corrosion is the corrosion that occurs when metals or alloys react with liquids, such as water, acids, bases, salts, etc. Wet corrosion can result in the formation of hydroxides, sulfates, chlorides, etc. on the surface of the metal or alloy. Wet corrosion can be prevented or reduced by applying protective coatings, such as paints, varnishes, etc., or by using cathodic protection, which is the technique of making the metal or alloy the cathode of an electrochemical cell, and supplying electrons to it from an external source, such as a sacrificial anode or an impressed current.

Chemistry of Engineering Materials: Cement

Cement is a binding material that is used in construction and civil engineering. It is composed of various chemical compounds that react with water to form a hard and durable mass. Cement can be classified into two main types based on the way it sets and hardens: hydraulic cement and non-hydraulic cement.

- Hydraulic cement hardens due to the addition of water. It can be used underwater or in moist conditions. The most common type of hydraulic cement is Portland cement, which is made up of four main compounds: tricalcium silicate ($3\text{CaO} \cdot \text{SiO}_2$), dicalcium silicate ($2\text{CaO} \cdot \text{SiO}_2$), tricalcium aluminate ($3\text{CaO} \cdot \text{Al}_2\text{O}_3$), and a tetra-calcium aluminoferrite

($4\text{CaO} \cdot \text{Al}_2\text{O}_3\text{Fe}_2\text{O}_3$) . These compounds undergo hydration reactions with water to form calcium silicate hydrate (C-S-H), calcium hydroxide ($\text{Ca}(\text{OH})_2$), and calcium aluminate hydrate (C-A-H), which are responsible for the strength and durability of cement.

- Non-hydraulic cement hardens by carbonation with the carbon dioxide present in the air. It cannot be used underwater or in moist conditions. The most common type of non-hydraulic cement is lime, which is made of calcium oxide (CaO) or calcium hydroxide ($\text{Ca}(\text{OH})_2$). When lime is exposed to air, it reacts with carbon dioxide to form calcium carbonate (CaCO_3), which binds the aggregates together.

Cement is usually mixed with sand, crushed stone, or other inert materials to form mortar or concrete. Mortar is used to join bricks, stones, or tiles together, while concrete is used to make foundations, walls, floors, or other structures. The properties of mortar and concrete depend on the type and proportion of cement, water, and aggregates used.

4. To enable the students to understand and apply detailed concepts of water source, water impurities, hardness

- Water source is the origin of water that is used for various purposes such as drinking, irrigation, industry, etc. Water sources can be classified into surface water and groundwater.
- Surface water is the water that collects on the land surface, such as rivers, lakes, reservoirs, etc. Surface water is influenced by precipitation, evaporation, runoff, and human activities.
- Groundwater is the water that exists below the land surface, in the pores and cracks of rocks and soil. Groundwater is replenished by infiltration of rainwater and snowmelt, and discharged by springs, wells, and seepage to surface water bodies.
- Water impurities are the substances that are present in water and affect its quality, quantity, or suitability for use. Water impurities can be classified into physical, chemical, and biological impurities.
- Physical impurities are the suspended or dissolved solids, such as sand, clay, silt, organic matter, etc. that affect the appearance, taste, odor, and turbidity of water.
- Chemical impurities are the dissolved or undissolved substances, such as salts, metals, acids, bases, gases, etc. that affect the pH, hardness, conductivity, and corrosivity of water.
- Biological impurities are the living or dead organisms, such as bacteria, viruses, algae, fungi, protozoa, etc. that affect the health, safety, and aesthetic quality of water.
- Hardness is the property of water that causes it to form scales or deposits

on surfaces, such as pipes, boilers, heaters, etc. Hardness is mainly caused by the presence of calcium and magnesium ions in water, which react with soap to form insoluble precipitates.

- Hardness can be classified into temporary hardness and permanent hardness.
- Temporary hardness is the hardness that can be removed by boiling or adding lime to water. Temporary hardness is caused by the presence of bicarbonates of calcium and magnesium in water, which decompose to form carbonates upon heating or reacting with lime.
- Permanent hardness is the hardness that cannot be removed by boiling or adding lime to water. Permanent hardness is caused by the presence of chlorides, sulfates, and nitrates of calcium and magnesium in water, which do not decompose upon heating or reacting with lime.

Water and Boiler Troubles Used in Industry

Water is a vital medium for many industrial and commercial processes, such as boilers, cooling towers, cooling systems and other closed circuit systems. However, water can also cause various problems in these systems, such as corrosion, scaling, fouling, and microbiological growth. These problems can affect the efficiency, safety, and reliability of the equipment, and may require costly repairs or replacements. Therefore, water treatment and quality control are essential for preventing and solving these problems.

Some of the common water and boiler troubles used in industry are:

- **Priming:** This is the phenomenon of water droplets being carried away with the steam from the boiler, resulting in wet and low-quality steam. Priming can be caused by high water levels, sudden changes in steam demand, improper boiler design, or contaminated feed water. Priming can lead to water hammer, erosion, and damage to the steam system .
- **Foaming:** This is the formation of a layer of froth or stable foam on the surface of the water in the boiler, which reduces the steam space and increases the water level. Foaming can be caused by high concentrations of dissolved solids, organic matter, or oil in the feed water, or by improper boiler operation. Foaming can result in carryover, corrosion, and reduced efficiency.
- **Scaling:** This is the deposition of hard and insoluble minerals, such as calcium carbonate, on the internal surfaces of the boiler and the heat transfer equipment. Scaling can be caused by high water hardness, high temperatures, or low pH. Scaling can reduce the heat transfer efficiency, increase the fuel consumption, and cause overheating and failure of the equipment .
- **Corrosion:** This is the deterioration of the metal surfaces of the boiler and the steam system due to chemical or electrochemical reactions with the water or the steam. Corrosion can be caused by low pH, dissolved oxy-

gen, carbon dioxide, chlorides, sulfates, or microbiological activity. Corrosion can result in leaks, pitting, cracking, and reduced strength of the equipment .

- **Fouling:** This is the accumulation of soft and sticky substances, such as oil, grease, organic matter, or microbiological growth, on the internal surfaces of the boiler and the heat transfer equipment. Fouling can be caused by contaminated feed water, poor water treatment, or inadequate cleaning. Fouling can reduce the heat transfer efficiency, increase the pressure drop, and cause blockages and malfunctions of the equipment .

To prevent and solve these water and boiler troubles, various water treatment methods and techniques are used, such as:

- **Pre-treatment:** This is the process of removing or reducing the impurities and contaminants in the feed water before entering the boiler, such as hardness, dissolved solids, organic matter, oil, and gases. Pre-treatment methods include filtration, softening, deaeration, demineralization, and reverse osmosis .
- **Internal treatment:** This is the process of adding chemicals or agents to the water inside the boiler to control the water quality and prevent or reduce the problems, such as scaling, corrosion, foaming, and priming. Internal treatment methods include pH adjustment, oxygen scavenging, alkalinity control, sludge conditioning, and antifoaming .
- **External treatment:** This is the process of treating the water outside the boiler after it has been used for steam generation or heat transfer, such as condensate, blowdown, or cooling water. External treatment methods include neutralization, dealkalization, desalination, and disinfection .
- **Monitoring and control:** This is the process of measuring and regulating the water quality and the operating parameters of the boiler and the steam system, such as temperature, pressure, flow, pH, conductivity, dissolved oxygen, and corrosion rate. Monitoring and control methods include sensors, meters, indicators, controllers, and alarms .

By applying these water treatment methods and techniques, the water and boiler troubles can be prevented or solved, and the performance, safety, and reliability of the industrial and commercial processes can be improved.

5. To enable the students to understand detailed concepts related to polymers, Polymerization, Polymer Blends

- Polymers are large molecules composed of repeating units called monomers, which are linked by covalent bonds.
- Polymerization is the process of forming polymers from monomers, either by addition or condensation reactions.

- Addition polymerization involves the joining of monomers without the elimination of any atoms or molecules, such as water. The monomers have double or triple bonds that open up and form new bonds with other monomers. For example, ethene can polymerize to form polyethylene.
- Condensation polymerization involves the joining of monomers with the elimination of a small molecule, such as water, ammonia, or hydrogen chloride. The monomers have functional groups that react with each other to form new bonds and release the by-product. For example, ethylene glycol and terephthalic acid can polymerize to form polyethylene terephthalate (PET) and water.
- Polymer blends are mixtures of two or more polymers that are physically mixed but not chemically bonded. They can be homogeneous or heterogeneous, depending on the compatibility and miscibility of the polymers. Polymer blends can have improved properties compared to the individual polymers, such as strength, toughness, thermal stability, or biodegradability. For example, polystyrene and polybutadiene can form a blend that has better impact resistance than pure polystyrene.

Polymer Composites

- A polymer composite is a multi-phase material in which reinforcing fillers are integrated with a polymer matrix, resulting in synergistic mechanical properties that cannot be achieved from either component alone.
- A polymer is a natural or synthetic substance composed of very large molecules, called macromolecules, that are multiples of simpler chemical units called monomers.
- A polymer composite can consist of a continuous phase or matrix, which is typically the resin or polymer phase, and a dispersed phase or reinforcement, which can be continuous or noncontinuous, and can be made of various materials such as fibers, particles, or flakes .
- Polymer composites are increasingly being used in various engineering fields, such as aerospace, automotive, biomedical, construction, and energy, because of their advantages over conventional materials, such as high strength-to-weight ratio, corrosion resistance, design flexibility, and low cost .
- Polymer composites can be classified into different types based on the nature of the matrix, the reinforcement, and the interface between them. Some common types of polymer composites are:
 - Thermoset composites: These composites have a matrix that is a thermosetting polymer, which undergoes an irreversible chemical reaction during curing and cannot be remelted or reshaped. Examples of thermoset composites are epoxy composites, polyester composites, and vinyl ester composites.
 - Thermoplastic composites: These composites have a matrix that is a thermoplastic polymer, which can be melted and reshaped by heating

and cooling. Examples of thermoplastic composites are polypropylene composites, nylon composites, and polycarbonate composites.

- Fiber-reinforced composites: These composites have a matrix that is reinforced by fibers, which can be continuous or discontinuous, and can be made of various materials such as glass, carbon, aramid, or natural fibers. Examples of fiber-reinforced composites are glass fiber-reinforced polymer (GFRP), carbon fiber-reinforced polymer (CFRP), and natural fiber-reinforced polymer (NFRP).
- Particle-reinforced composites: These composites have a matrix that is reinforced by particles, which can be spherical, cubic, or irregular in shape, and can be made of various materials such as ceramics, metals, or polymers. Examples of particle-reinforced composites are polymer nanocomposites, metal matrix composites, and ceramic matrix composites.
- Hybrid composites: These composites have a matrix that is reinforced by two or more types of reinforcements, which can have different properties and functions. Examples of hybrid composites are carbon-glass hybrid composites, metal-polymer hybrid composites, and ceramic-polymer hybrid composites.

Content Contact

- Content contact is a term used in web design and development to describe the process of creating, editing, and updating the content of a website or an application.
- Content contact involves the collaboration between the content creators, such as writers, editors, designers, and developers, and the content owners, such as clients, stakeholders, or users, who have the authority and responsibility to approve and publish the content.
- Content contact aims to ensure that the content is accurate, relevant, consistent, engaging, and aligned with the goals and needs of the content owners and the target audience.
- Content contact can be divided into four phases: content planning, content creation, content review, and content publishing.
 - Content planning involves defining the content strategy, scope, structure, and schedule for the website or the application.
 - Content creation involves producing the content according to the content plan, using various tools and formats, such as text, images, videos, audio, etc.
 - Content review involves checking the quality, accuracy, and suitability of the content, as well as obtaining feedback and approval from the content owners and other stakeholders.
 - Content publishing involves uploading, updating, and managing the content on the website or the application, using various platforms and systems, such as content management systems (CMS), web servers, etc.

- Content contact requires effective communication, coordination, and documentation among the content contact team members, as well as the use of best practices and standards for content creation and management.

Hours

- An hour is a unit of time that is equal to 60 minutes or 3,600 seconds.
- The symbol for hour is h or hr.
- Hours are commonly used to measure the duration of events, activities, or processes.
- Hours are also used to indicate the time of day in various systems, such as the 12-hour clock or the 24-hour clock.
- The 12-hour clock divides the day into two periods of 12 hours each, called AM (ante meridiem, meaning before noon) and PM (post meridiem, meaning after noon).
- The 24-hour clock uses a single cycle of 24 hours to represent the time of day, starting from midnight (00:00) and ending at midnight (24:00) of the next day.
- The 24-hour clock is also known as the military time or the international standard time.
- The origin of the hour as a unit of time can be traced back to ancient civilizations, such as the Babylonians, Egyptians, Greeks, and Romans, who divided the day into 12 or 24 equal parts based on the movement of the sun and the stars.
- The modern definition of the hour as 60 minutes was established by the International System of Units (SI) in 1960.

Unit-1: 8

- Unit-1: 8 is a topic that covers the basics of binary arithmetic and logic operations.
- Binary arithmetic is the process of performing mathematical operations on binary numbers, which are numbers composed of only two digits: 0 and 1.
- Logic operations are the process of applying logical rules to binary values, such as AND, OR, NOT, XOR, etc.
- Some of the subtopics covered in Unit-1: 8 are:
 - Binary addition and subtraction: how to add and subtract binary numbers using the rules of carry and borrow.
 - Binary multiplication and division: how to multiply and divide binary numbers using the rules of shift and add or subtract.
 - Binary complement: how to find the inverse of a binary number by flipping all the bits and adding 1.

- Binary codes: how to represent different types of data using binary digits, such as decimal, hexadecimal, octal, ASCII, etc.
- Logic gates: how to implement basic logic operations using electronic circuits that have two or more inputs and one output.
- Truth tables: how to represent the output of a logic gate or a combination of logic gates for all possible input values.
- Boolean algebra: how to simplify and manipulate logic expressions using the rules and laws of Boolean logic, such as De Morgan's laws, distributive law, etc.
- Karnaugh maps: how to minimize logic expressions using a graphical method that groups adjacent cells with the same output value.

Atomic and Molecular Structure: Molecular orbital's of diatomic molecules, Bond Order

- Molecular orbital theory (MOT) is a method of describing the electronic structure of molecules using quantum mechanics. It assumes that the atomic orbitals of the atoms in a molecule can combine to form molecular orbitals that belong to the whole molecule.
- Molecular orbitals can be either bonding or antibonding, depending on whether they have lower or higher energy than the original atomic orbitals. Bonding orbitals are more stable and tend to be occupied by electrons, while antibonding orbitals are less stable and tend to be empty.
- The bond order of a molecule is a measure of the number of bonds between two atoms. It can be calculated using the formula:

$$\text{bond order} = 1/2 (\text{n bonding} - \text{n antibonding})$$

where n bonding is the number of electrons in the bonding molecular orbitals, and n antibonding is the number of electrons in the antibonding molecular orbitals. The bond order is a metric that indicates how strong a bond is .

- Diatomic molecules are molecules that consist of two atoms of the same or different elements. For example, H₂, O₂, N₂, CO, and NO are diatomic molecules.
- To apply molecular orbital theory to diatomic molecules, we need to consider the overlap of the atomic orbitals of the two atoms. The type and number of molecular orbitals that are formed depend on the symmetry and energy of the atomic orbitals.
- For homonuclear diatomic molecules, which are composed of two identical atoms, the atomic orbitals have the same energy and symmetry. Therefore, they can combine to form two molecular orbitals of equal energy, one bonding and one antibonding. For example, the 1s orbitals of two

hydrogen atoms can form a sigma (σ) bonding orbital and a sigma star (σ^*) antibonding orbital. The bond order of H₂ is 1, since it has two electrons in the bonding orbital and none in the antibonding orbital.

- For heteronuclear diatomic molecules, which are composed of two different atoms, the atomic orbitals have different energies and symmetries. Therefore, they can combine to form two molecular orbitals of unequal energy, one bonding and one antibonding. The bonding orbital is closer in energy to the lower-energy atomic orbital, while the antibonding orbital is closer in energy to the higher-energy atomic orbital. For example, the 2s orbital of carbon and the 2p orbital of oxygen can form a sigma (σ) bonding orbital and a sigma star (σ^*) antibonding orbital. The bond order of CO is 3, since it has eight electrons in the bonding orbitals and two electrons in the antibonding orbitals.
- For diatomic molecules with more than one pair of atomic orbitals, such as O₂, N₂, and F₂, we need to consider the splitting schemes for the 2s and 2p orbitals. The 2s orbitals can form a sigma (σ) bonding orbital and a sigma star (σ^*) antibonding orbital, while the 2p orbitals can form two pi (π) bonding orbitals, two pi star (π^*) antibonding orbitals, and one sigma (σ) bonding orbital and one sigma star (σ^*) antibonding orbital. The order of the molecular orbitals depends on the relative energies of the 2s and 2p orbitals of the atoms. For O₂ and F₂, the 2s orbitals are lower in energy than the 2p orbitals, so the sigma (σ) bonding orbital is below the pi (π) bonding orbitals. For N₂, the 2s orbitals are higher in energy than the 2p orbitals, so the sigma (σ) bonding orbital is above the pi (π) bonding orbitals.

Magnetic Characteristics and Numerical Problems

- Magnetic characteristics are the properties of a material that determine its behavior in a magnetic field, such as magnetic permeability, magnetic susceptibility, magnetic moment, magnetic polarization, etc.
- Magnetic permeability is the measure of how easily a material can be magnetized by an external magnetic field. It is defined as the ratio of the magnetic flux density (B) to the magnetic field strength (H) in a material: $\mu = B/H$. The SI unit of magnetic permeability is henry per meter (H/m).
- Magnetic susceptibility is the measure of how much a material is magnetized by an applied magnetic field. It is defined as the ratio of the magnetic moment (M) per unit volume to the applied magnetic field (H): $\chi = M/H$. The SI unit of magnetic susceptibility is dimensionless, but it is often expressed in parts per million (ppm) or emu/cm³.
- Magnetic moment is the measure of the strength and direction of a magnet

or a current loop. It is defined as the product of the current (I) and the area (A) of the loop: $m = IA$. The SI unit of magnetic moment is ampere-meter squared ($A \cdot m^2$).

- Magnetic polarization is the measure of the density of magnetic moments in a material. It is defined as the magnetic moment (M) per unit volume: $J = M/V$. The SI unit of magnetic polarization is ampere per meter (A/m).
- Numerical problems are the problems that involve calculations using the formulas and concepts of magnetic characteristics. They usually require the knowledge of the values of the magnetic characteristics of different materials, such as air, iron, steel, etc.
- Some examples of numerical problems are:
 - Calculate the magnetic flux density (B) in a magnetic circuit that consists of an iron core with a cross-sectional area of $0.01 m^2$ and a length of $0.5 m$, and a coil with 1000 turns and a current of $2 A$. Assume that the relative permeability of iron is 5000 and the permeability of free space is $4 \times 10^{-7} H/m$.
 - Solution: The magnetic flux density (B) is given by the formula: $B = \mu NI/l$, where μ is the magnetic permeability, N is the number of turns, I is the current, and l is the length of the core. The magnetic permeability of iron is given by: $\mu = \mu_r \mu_0$, where μ_r is the relative permeability and μ_0 is the permeability of free space. Substituting the values, we get: $B = (5000)(4 \times 10^{-7})(1000)(2)/(0.5) = 0.628 T$.
 - Calculate the magnetic susceptibility (χ) of a material that has a magnetic polarization (J) of $0.01 A/m$ when placed in a magnetic field (H) of $0.1 A/m$.
 - Solution: The magnetic susceptibility (χ) is given by the formula: $\chi = J/H$. Substituting the values, we get: $\chi = 0.01/0.1 = 0.1$.
 - Calculate the magnetic moment (m) of a current loop that has a current (I) of $0.5 A$ and an area (A) of $0.02 m^2$.
 - Solution: The magnetic moment (m) is given by the formula: $m = IA$. Substituting the values, we get: $m = (0.5)(0.02) = 0.01 A \cdot m^2$.

Chemistry of Advanced Materials

- Advanced materials are materials that have novel properties and functions that are superior to conventional materials.
- Advanced materials can be classified into three main categories: nanomaterials, biomaterials and energy materials.
- Nanomaterials are materials that have at least one dimension in the nanometer range ($1-100 nm$). They exhibit unique physical, chemical and

biological properties due to their small size and high surface area.

- Biomaterials are materials that interact with biological systems, such as tissues, cells and organs. They can be used for medical applications, such as implants, drug delivery and tissue engineering.
- Energy materials are materials that can convert, store or transport energy, such as solar cells, batteries, fuel cells and supercapacitors. They can help address the global challenges of energy supply and demand.
- The chemistry of advanced materials involves the design, synthesis, characterization and modification of materials at the molecular level, as well as the understanding of their structure-property relationships.
- The chemistry of advanced materials requires the integration of knowledge and techniques from different disciplines, such as organic, inorganic, physical, analytical and computational chemistry, as well as physics, biology and engineering.
- The chemistry of advanced materials has a wide range of applications in various fields, such as electronics, optics, catalysis, sensors, biotechnology, medicine and environmental engineering.
- The chemistry of advanced materials is a dynamic and rapidly evolving field that offers many opportunities for innovation and discovery.

Liquid Crystals; Introduction, Types and Applications of liquid crystals, Industrially

- Liquid crystals are substances that exhibit a phase of matter that is between a solid and a liquid, with some degree of molecular order and mobility .
- Liquid crystals can be classified into two main categories: **thermotropic** and **lyotropic** .
 - Thermotropic liquid crystals are formed by heating a crystalline solid and lose their order when heated above a certain temperature .
 - Lyotropic liquid crystals are formed by dissolving certain molecules in a solvent and depend on the concentration and temperature of the solution .
- Liquid crystals can also be classified by the type of molecular order they exhibit, such as **nematic**, **smectic**, **cholesteric**, **discotic**, and **columnar** .
 - Nematic liquid crystals have molecules that are aligned along a common direction, called the director, but have no positional order .
 - Smectic liquid crystals have molecules that are arranged in layers, with each layer having a different director .
 - Cholesteric liquid crystals are a type of nematic liquid crystals that have a helical twist of the director along the axis of the layers .
 - Discotic liquid crystals have molecules that are shaped like discs and are arranged in columns .

- Columnar liquid crystals have molecules that are shaped like rods and are arranged in columns .
- Liquid crystals have many applications in various fields, such as display devices, sensors, lasers, medical diagnostics, and pharmaceuticals .
 - Display devices use liquid crystals to modulate the light passing through them, creating images on screens .
 - Sensors use liquid crystals to detect changes in temperature, pressure, electric field, magnetic field, or chemical environment, by measuring the changes in their optical properties .
 - Lasers use liquid crystals to control the polarization, wavelength, and intensity of the laser beam, by applying electric or magnetic fields .
 - Medical diagnostics use liquid crystals to monitor the health status of patients, such as blood glucose level, cholesterol level, or infection, by using liquid crystal-based biosensors .
 - Pharmaceuticals use liquid crystals to deliver drugs to specific targets in the body, such as tumors, by using liquid crystal-based carriers or vesicles .

Important Materials Used as Liquid Crystals

- Liquid crystals are substances that exhibit properties of both liquids and crystalline solids, such as fluidity, anisotropy, and birefringence.
- Liquid crystals can be classified into two main types: thermotropic and lyotropic.
- Thermotropic liquid crystals are temperature-dependent and change their phase and orientation according to the external temperature.
- Lyotropic liquid crystals are concentration-dependent and change their phase and orientation according to the amount of solvent (usually water) present in the system.
- Some examples of thermotropic liquid crystals are:
 - High molar mass materials, such as polymers, that can form liquid crystalline phases by aligning their long and flexible chains.
 - Low molar mass materials, such as aromatic compounds, that can form liquid crystalline phases by stacking their rigid and flat molecules.
- Some examples of lyotropic liquid crystals are:
 - Amphiphilic molecules, such as long-chain alcohols, surfactants, and phospholipids, that can form liquid crystalline phases by arranging their hydrophilic and hydrophobic parts in different ways.
 - Biological materials, such as DNA, gelatin, and tobacco mosaic virus, that can form liquid crystalline phases by self-assembling their rodlike or helical structures.
- Liquid crystals have many applications in various fields, such as display technology, optical devices, sensors, and biomedicine.

Graphite and Fullerene; Introduction, Structure and Applications

- Graphite and fullerene are two forms of carbon that have different structures and properties.
- Graphite has a layered structure in which hexagonal rings of carbon atoms are connected to each other by weak intermolecular forces. Graphite is a good conductor of electricity and heat, and has a high melting point and lubricating property.
- Fullerene is a class of molecules that have a spherical, ellipsoidal, or cylindrical shape. The most common fullerene is C₆₀, also known as buckminsterfullerene or buckyball, which has 60 carbon atoms arranged in a soccer ball-like structure. Fullerene can also have other numbers of carbon atoms, such as C₇₀, C₇₆, C₈₄, etc. Fullerene has a high symmetry and stability, and can form complexes with other atoms or molecules.
- Graphene is a single layer of graphite, which is one atom thick. Graphene has a honeycomb lattice structure and has extraordinary mechanical, electrical, optical, and thermal properties. Graphene is considered as a wonder material for various applications in nanotechnology, electronics, biomedicine, and energy.
- Some of the applications of graphite are:
 - As electrodes in batteries, fuel cells, and electrochemical sensors.
 - As lubricants and additives in polymers and composites.
 - As catalysts and catalyst supports in chemical reactions.
 - As moderators and reflectors in nuclear reactors.
- Some of the applications of fullerene are:
 - As drug delivery agents, contrast agents, and biosensors in biomedicine.
 - As photovoltaic materials, light-emitting diodes, and transistors in optoelectronics.
 - As hydrogen storage materials, catalysts, and superconductors in energy and nanotechnology.
 - As pigments, coatings, and additives in cosmetics and paints.
- Some of the applications of graphene are:
 - As transparent electrodes, touch screens, and flexible displays in electronics.
 - As membranes, filters, and desalination devices in water purification.
 - As sensors, biosensors, and nanoprobe in biotechnology and nanotechnology.
 - As supercapacitors, batteries, and solar cells in energy storage and conversion.

Nanomaterials: Introduction, Preparation, Characteristics and Applications

Nanomaterials are materials that have at least one dimension in the range of 1 to 100 nanometers (nm). They exhibit unique optical, magnetic, electrical, and other properties that are different from those of bulk materials. Nanomaterials can be classified into two types: nanostructured and nanoparticle materials. Nanostructured materials have a regular pattern or structure at the nanoscale, such as nanotubes, nanowires, nanosheets, etc. Nanoparticle materials have no specific structure or shape at the nanoscale, such as quantum dots, nanocrystals, nanoclusters, etc.

Preparation of Nanomaterials

There are two main approaches for preparing nanomaterials: top-down and bottom-up methods. Top-down methods involve breaking down larger materials into smaller ones using physical or chemical processes, such as milling, etching, lithography, etc. Bottom-up methods involve building up nanomaterials from smaller units using chemical or biological processes, such as self-assembly, precipitation, sol-gel, etc.

Characteristics of Nanomaterials

Some of the common characteristics of nanomaterials are:

- High surface area to volume ratio: Nanomaterials have a large fraction of atoms or molecules on their surface, which can enhance their reactivity, catalytic activity, adsorption, etc.
- Quantum confinement: Nanomaterials have discrete energy levels due to the confinement of electrons in a small space, which can affect their optical, electronic, and magnetic properties.
- Size and shape dependence: Nanomaterials have different properties depending on their size and shape, which can be controlled by varying the synthesis conditions and functionalization methods.
- Surface plasmon resonance: Nanomaterials can interact with electromagnetic waves and generate collective oscillations of electrons on their surface, which can result in enhanced scattering, absorption, and emission of light.

Applications of Nanomaterials

Nanomaterials have a wide range of applications in various fields, such as:

- Electronics: Nanomaterials can be used to fabricate nanoscale devices, such as transistors, sensors, memory, displays, etc., that can improve the performance, efficiency, and functionality of electronic systems.

- **Medicine:** Nanomaterials can be used to deliver drugs, genes, or imaging agents to specific targets in the body, as well as to diagnose, treat, and monitor diseases, such as cancer, infections, inflammation, etc.
- **Energy:** Nanomaterials can be used to enhance the conversion, storage, and utilization of energy, such as solar cells, batteries, fuel cells, supercapacitors, etc., that can offer higher efficiency, capacity, and stability.
- **Environment:** Nanomaterials can be used to remove pollutants, pathogens, or toxins from water, air, or soil, as well as to detect and monitor environmental quality, such as sensors, filters, catalysts, etc., that can offer high sensitivity, selectivity, and durability.

Nanomaterials, Carbon Nanotubes (CNT)

- Nanomaterials are materials that have at least one dimension in the nanometer range (1-100 nm).
- Carbon nanomaterials are a family of carbon allotropes, consisting of different forms and shapes of carbon atoms arranged in various structures.
- Carbon nanotubes (CNTs) are one of the most important types of carbon nanomaterials. They are cylindrical tubes of carbon atoms, with diameters ranging from a few nanometers to tens of nanometers, and lengths up to several micrometers or even millimeters.
- CNTs can be classified into two main categories: single-walled carbon nanotubes (SWCNTs) and multi-walled carbon nanotubes (MWCNTs).
 - SWCNTs consist of a single layer of carbon atoms arranged in a hexagonal lattice, forming a seamless cylinder of graphene.
 - MWCNTs consist of multiple concentric layers of carbon atoms, forming nested tubes of different diameters.
- CNTs have remarkable physical and chemical properties, such as high strength, stiffness, elasticity, thermal conductivity, electrical conductivity, and optical activity. They also have potential applications in various fields, such as nanoelectronics, nanomedicine, nanocomposites, energy storage, and sensors.
- CNTs are mainly synthesized by chemical vapor deposition (CVD), which involves the decomposition of a carbon-containing gas (such as methane, acetylene, or carbon monoxide) on a metal catalyst (such as iron, nickel, or cobalt) at high temperatures (600-1200 °C).
- CNTs can also be functionalized or modified by attaching different molecules or groups to their surface or ends, to enhance their solubility, dispersibility, biocompatibility, or reactivity. Some common methods of functionalization include oxidation, reduction, covalent bonding, non-covalent bonding, and polymer wrapping.

Green Chemistry: Introduction, 12 principles and importance of green synthesis, Green

- Green chemistry, also called sustainable chemistry or circular chemistry, is an area of chemistry and chemical engineering focused on the design of products and processes that minimize or eliminate the use and generation of hazardous substances .
- Green chemistry applies across the life cycle of a chemical product, including its design, manufacture, use, and ultimate disposal.
- Green chemistry can also be defined through the use of metrics, such as mass, energy, hazardous substance reduction or elimination, and life cycle environmental impacts.
- Green chemistry is based on 12 principles that provide guidelines for the design of greener products and processes. These principles are:
 1. Prevent waste: Design chemical syntheses to prevent waste, leaving no waste to treat or clean up.
 2. Maximize atom economy: Design syntheses so that the final product contains the maximum proportion of the starting materials. There should be few, if any, wasted atoms.
 3. Design less hazardous chemical syntheses: Design syntheses to use and generate substances with little or no toxicity to humans and the environment.
 4. Design safer chemicals and products: Design chemical products to be fully effective, yet have little or no toxicity.
 5. Use safer solvents and reaction conditions: Avoid using solvents, separation agents, or other auxiliary chemicals. If these chemicals are necessary, use innocuous chemicals.
 6. Increase energy efficiency: Run chemical reactions at ambient temperature and pressure whenever possible.
 7. Use renewable feedstocks: Use starting materials (also known as feedstocks) that are renewable rather than depleting. Renewable feedstocks are often made from agricultural products or are the wastes of other processes; depleting feedstocks are made from fossil fuels (petroleum, natural gas, or coal) or are mined.
 8. Avoid chemical derivatives: Avoid using blocking or protecting groups or any temporary modifications if possible. Derivatives use additional reagents and generate waste.
 9. Use catalysts, not stoichiometric reagents: Minimize waste by using catalytic reactions. Catalysts are used in small amounts and can carry out a single reaction many times. They are preferable to stoichiometric reagents, which are used in excess and work only once.
 10. Design chemicals and products to degrade after use: Design chemical products to break down to innocuous substances after use so that

they do not accumulate in the environment.

11. Analyze in real time to prevent pollution: Include in-process real-time monitoring and control during syntheses to minimize or eliminate the formation of byproducts.
 12. Minimize the potential for accidents: Design chemicals and their forms (solid, liquid, or gas) to minimize the potential for chemical accidents including explosions, fires, and releases to the environment.
- The importance of green chemistry lies in its potential to reduce or eliminate the environmental and health impacts of chemical products and processes, while maintaining or enhancing their performance and economic viability .
 - Green chemistry can also provide benefits such as energy and resource savings, improved competitiveness and innovation, reduced liability and risk, and enhanced public image and reputation .
 - Green synthesis, also known as eco-friendly synthesis or sustainable synthesis, is a branch of green chemistry that aims to develop synthetic methods that are environmentally benign, economically feasible, and socially acceptable.
 - Green synthesis can involve the use of green solvents, green catalysts, green reagents, green energy sources, and green reaction conditions.
 - Green synthesis can also involve the use of natural or biological sources, such as plants, microorganisms, enzymes, or biomimetic systems, to produce or modify chemical compounds.
 - Green synthesis can offer advantages such as reduced waste generation, improved atom economy, enhanced selectivity and specificity, reduced toxicity and hazard, and increased biodegradability and biocompatibility.

Chemicals, Synthesis of typical organic compounds by conventional and Green route (Adipic Acid)

- Adipic acid is a dicarboxylic acid with the formula $\text{HOOC}(\text{CH}_2)_4\text{COOH}$. It is mainly used as a precursor for the production of nylon 6,6.
- The conventional route of synthesis of adipic acid involves the oxidation of benzene or cyclohexane with nitric acid or air, followed by hydrolysis of the resulting cyclohexanone oxime.
- The conventional route has several drawbacks, such as high energy consumption, high pressure and temperature requirements, and the generation of large amounts of nitrous oxide, a greenhouse gas and an ozone depleter.

- The green route of synthesis of adipic acid aims to use renewable biomass as the starting material, such as glucose, and to employ milder and more environmentally friendly reaction conditions and catalysts.
- One example of the green route is the two-step transformation of glucose into adipic acid via glucaric acid, as reported by Liu et al. .
- In the first step, glucose is oxidized to glucaric acid by carbon nanotube-supported platinum nanoparticles in water under mild conditions (80 °C, 1 atm O₂).
- In the second step, glucaric acid is dehydrated to adipic acid by a solid acid catalyst, such as Amberlyst-15, at 140 °C.
- The green route has several advantages, such as high atom economy, high selectivity, low energy consumption, and low environmental impact.

Acid and Paracetamol

- Paracetamol is a drug that belongs to the class of analgesics (pain relievers) and antipyretics (fever reducers).
- Paracetamol is also known as acetaminophen in some countries, such as the United States and Canada.
- Paracetamol can be combined with other drugs, such as mefenamic acid or ascorbic acid, to enhance its effects or to treat different conditions.
- Paracetamol, being an acidic drug, is best absorbed in an acidic environment, such as the stomach.
- Paracetamol can cause side effects, such as nausea, vomiting, rash, liver damage, and allergic reactions, especially if taken in high doses or for a long time.
- Paracetamol can also interact with some medications, such as anticoagulants, anticonvulsants, alcohol, and some antibiotics, and increase the risk of bleeding, seizures, liver toxicity, or stomach irritation.
- Paracetamol should be used with caution in patients with liver or kidney problems, asthma, or blood disorders, and in pregnant or breastfeeding women.
- Paracetamol should be taken as directed by a doctor or pharmacist, and the recommended dose and duration should not be exceeded.

Environmental impact of Green chemistry on society

- Green chemistry is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances.
- Green chemistry aims to protect human health and the environment by applying the principles of sustainability, efficiency, and safety to chemistry.
- Green chemistry can have positive impacts on society, such as:
 - Reducing the amount of waste and pollution generated by chemical

industries and products, and thus saving costs and resources.

- Improving the quality and safety of consumer products, such as cosmetics, pharmaceuticals, food, and textiles, by using less toxic or renewable ingredients.
- Developing new and innovative solutions for environmental and social challenges, such as climate change, energy, water, and health.
- Creating new opportunities and markets for green products and technologies, and enhancing the competitiveness and profitability of chemical businesses.
- Educating and empowering the public and the scientific community to adopt greener practices and lifestyles, and to foster a culture of environmental responsibility and awareness.

Unit-2: 8

- This unit covers the topic of **data structures and algorithms**.
- Data structures are ways of organizing and storing data in a computer memory, such as arrays, lists, stacks, queues, trees, graphs, etc.
- Algorithms are step-by-step procedures for solving a problem or performing a task, such as sorting, searching, traversing, etc.
- Data structures and algorithms are closely related, as the choice of data structure affects the efficiency and complexity of the algorithm, and vice versa.
- The main goals of data structures and algorithms are to **optimize the time and space** required for solving a problem, as well as the **correctness and robustness** of the solution.
- Some common concepts and techniques in data structures and algorithms are:
 - **Abstract data types (ADTs)**: A way of defining the behavior and operations of a data structure without specifying its implementation details. Examples are stack, queue, set, map, etc.
 - **Recursion**: A way of solving a problem by breaking it down into smaller subproblems that have the same structure as the original problem, and then combining the solutions of the subproblems. Examples are factorial, Fibonacci, binary search, etc.
 - **Divide and conquer**: A way of solving a problem by dividing it into two or more subproblems that are easier to solve, and then combining the solutions of the subproblems. Examples are merge sort, quick sort, binary search tree, etc.
 - **Dynamic programming**: A way of solving a problem by storing and reusing the solutions of overlapping subproblems, instead of re-computing them. Examples are matrix chain multiplication, longest common subsequence, knapsack problem, etc.
 - **Greedy algorithm**: A way of solving a problem by making the locally optimal choice at each step, without considering the global consequences. Examples are activity selection, Huffman coding, Di-

- ijkstra's algorithm, etc.
- **Backtracking:** A way of solving a problem by exploring all possible solutions and discarding the ones that do not satisfy the constraints. Examples are n-queens problem, sudoku, Hamiltonian cycle, etc.
 - **Branch and bound:** A way of solving a problem by pruning the search space based on some criteria, such as lower and upper bounds, heuristic functions, etc. Examples are travelling salesman problem, 0/1 knapsack problem, etc.
 - **Graph algorithms:** A way of solving problems that involve nodes and edges, such as finding the shortest path, the minimum spanning tree, the maximum flow, the connected components, etc. Examples are breadth-first search, depth-first search, Prim's algorithm, Kruskal's algorithm, Ford-Fulkerson algorithm, etc.
 - **Sorting algorithms:** A way of arranging the elements of a sequence in a specific order, such as ascending, descending, alphabetical, etc. Examples are bubble sort, insertion sort, selection sort, heap sort, merge sort, quick sort, radix sort, etc.
 - **Searching algorithms:** A way of finding an element or a set of elements that satisfy some criteria in a sequence or a data structure, such as linear search, binary search, interpolation search, hash table, binary search tree, etc.
 - **Complexity analysis:** A way of measuring the performance and efficiency of an algorithm in terms of the time and space required as a function of the input size. Examples are big-O notation, big-Omega notation, big-Theta notation, etc.

Spectroscopic Techniques and Applications: Elementary idea and simple applications of

Spectroscopy is the study of the interaction of matter with electromagnetic radiation. It can be used to identify the structure, composition, and properties of different substances by analyzing the spectrum of light they emit or absorb. There are different types of spectroscopic techniques, each based on a different region of the electromagnetic spectrum and a different type of interaction. Some of the most common spectroscopic techniques and their applications are:

- **Ultraviolet-visible (UV-Vis) spectroscopy:** This technique uses light in the ultraviolet and visible regions of the spectrum (200-800 nm) to measure the absorption of molecules. It can be used to determine the concentration, purity, and identity of organic and inorganic compounds, as well as to study the electronic transitions and molecular orbitals of molecules. Some applications of UV-Vis spectroscopy are:
 - Quantitative analysis of metal ions, organic dyes, drugs, and biomolecules in solution.

- Qualitative analysis of functional groups and chromophores in organic compounds.
- Kinetic studies of chemical reactions and enzyme catalysis.
- Molecular fingerprinting and forensic analysis.
- **Infrared (IR) spectroscopy:** This technique uses light in the infrared region of the spectrum (2.5-25 micrometers) to measure the vibration of molecular bonds. It can be used to identify the functional groups and molecular structure of organic and inorganic compounds, as well as to study the intermolecular interactions and hydrogen bonding of molecules. Some applications of IR spectroscopy are:
 - Structural elucidation and confirmation of organic and inorganic compounds.
 - Qualitative and quantitative analysis of gases, liquids, and solids.
 - Identification of polymers and plastics.
 - Detection of impurities and contaminants.
 - Monitoring of chemical reactions and processes.
- **Nuclear magnetic resonance (NMR) spectroscopy:** This technique uses radio waves in the presence of a magnetic field to measure the spin of atomic nuclei. It can be used to determine the structure, configuration, and environment of organic and inorganic compounds, as well as to study the dynamics and interactions of molecules. Some applications of NMR spectroscopy are:
 - Structural elucidation and confirmation of organic and inorganic compounds.
 - Determination of stereochemistry and conformation of molecules.
 - Analysis of complex mixtures and natural products.
 - Investigation of molecular motion and exchange processes.
 - Medical imaging and diagnosis (MRI).
- **Mass spectrometry (MS):** This technique uses an electric or magnetic field to separate and measure the mass-to-charge ratio of ionized molecules. It can be used to determine the molecular weight, formula, and structure of organic and inorganic compounds, as well as to study the fragmentation and rearrangement of molecules. Some applications of MS are:
 - Identification and characterization of organic and inorganic compounds.
 - Determination of isotopic composition and abundance of elements and molecules.
 - Analysis of trace and ultra-trace substances and contaminants.
 - Detection and quantification of drugs, metabolites, and biomolecules.
 - Proteomics and metabolomics studies.

UV, IR and NMR, Numerical problems

UV, IR and NMR are spectroscopic techniques that can be used to identify organic compounds by analyzing their molecular structure and functional groups.

UV spectroscopy

UV spectroscopy measures the absorption of ultraviolet light by a molecule. It can be used to determine the presence and extent of conjugation in a molecule, as well as the number and position of double bonds.

- The wavelength of maximum absorption (λ_{max}) is inversely proportional to the extent of conjugation. More conjugated molecules absorb at longer wavelengths (lower energy).
- The intensity of absorption (ϵ) is proportional to the number of double bonds in a molecule. More double bonds result in higher absorption coefficients.

Numerical problem

Example: A compound with molecular formula $\text{C}_6\text{H}_6\text{O}$ has a UV spectrum with $\lambda_{\text{max}} = 280 \text{ nm}$ and $\epsilon = 13000$. What is the structure of the compound?

Solution: The molecular formula indicates that the compound has 4 degrees of unsaturation, which means it has either 4 double bonds, 2 triple bonds, or 1 ring and 2 double bonds. The UV spectrum suggests that the compound has a high degree of conjugation, as the λ_{max} is relatively long and the ϵ is relatively high. Therefore, the most likely structure is a benzene ring with a hydroxyl group attached, as shown below.

Benzene ring with hydroxyl group

IR spectroscopy

IR spectroscopy measures the absorption of infrared light by a molecule. It can be used to determine the presence and type of functional groups in a molecule, as well as the bond strength and polarity.

- The frequency of absorption (ν) is proportional to the bond strength and inversely proportional to the bond length. Stronger and shorter bonds absorb at higher frequencies (higher energy).
- The intensity of absorption (I) is proportional to the bond polarity and the number of bonds. More polar and more numerous bonds result in higher absorption intensities.

Numerical problem

Example: A compound with molecular formula $\text{C}_4\text{H}_8\text{O}$ has an IR spectrum with a strong absorption at 1720 cm^{-1} and a weak absorption at 3300 cm^{-1} .

What is the structure of the compound?

Solution: The molecular formula indicates that the compound has 1 degree of unsaturation, which means it has either 1 double bond or 1 ring. The IR spectrum suggests that the compound has a carbonyl group, as the strong absorption at 1720 cm^{-1} corresponds to the C=O stretch. The weak absorption at 3300 cm^{-1} could be due to an alcohol group, as the O-H stretch is usually weak and broad. Therefore, the most likely structure is a cyclic ketone with a hydroxyl group attached, as shown below.

Cyclic ketone with hydroxyl group

NMR spectroscopy

NMR spectroscopy measures the absorption of radio waves by a molecule in a magnetic field. It can be used to determine the number and type of hydrogen atoms in a molecule, as well as their relative positions and environments.

- The chemical shift (δ) is the measure of how far a signal is from a reference signal, usually tetramethylsilane (TMS). It depends on the electronegativity and hybridization of the atoms attached to the hydrogen. More electronegative and more sp-hybridized atoms cause a higher chemical shift (more downfield).
- The integration (I) is the measure of the area under a signal. It is proportional to the number of hydrogen atoms that give rise to the signal. More hydrogen atoms result in a larger integration value.
- The multiplicity (M) is the measure of the splitting pattern of a signal. It depends on the number of neighboring hydrogen atoms that are not equivalent to the signal. The multiplicity follows the n+1 rule, where n is the number of neighboring hydrogen atoms. More neighboring hydrogen atoms result in a more complex multiplicity.

Numerical problem

Example: A compound with molecular formula $\text{C}_4\text{H}_8\text{O}$ has a ^1H NMR spectrum with three signals: a singlet at 2.1 ppm with integration of 3, a triplet at 1.2 ppm with integration of 3, and a quartet at 0.9 ppm with integration of 2. What is the structure of the

Stereochemistry: Optical isomerism in compounds without chiral carbon, Geometrical

- Stereochemistry is the branch of chemistry that studies the spatial arrangement of atoms and molecules and their effects on the physical and chemical properties of substances.

- Optical isomerism is a type of stereoisomerism that occurs when a compound has the same molecular formula and connectivity but different spatial orientation of atoms, resulting in non-superimposable mirror images.
- Optical isomers are also called enantiomers or chiral molecules. They have the property of rotating the plane of polarized light in opposite directions, which is called optical activity.
- Optical isomerism usually occurs in compounds that have one or more chiral centers, which are atoms that are bonded to four different groups. The most common example of a chiral center is a carbon atom with four different substituents.
- However, optical isomerism can also occur in compounds without chiral centers, such as allenes, biphenyls, spiranes, and some cyclic compounds. These compounds have a different type of chirality, which is called axial chirality, planar chirality, or helical chirality.
- Axial chirality occurs when a molecule has a twisted or helical shape around an axis, such as an allene or a biphenyl. Planar chirality occurs when a molecule has a plane of symmetry but not a center of symmetry, such as a metal complex with a square planar geometry. Helical chirality occurs when a molecule has a helical shape, such as a DNA strand.
- Geometrical isomerism is another type of stereoisomerism that occurs when a compound has the same molecular formula and connectivity but different spatial arrangement of atoms around a double bond, a ring, or a coordination complex.
- Geometrical isomers are also called cis-trans isomers or E-Z isomers. They have different physical and chemical properties, such as melting point, boiling point, solubility, and reactivity.
- Geometrical isomerism usually occurs in compounds that have restricted rotation around a bond or a ring, such as alkenes, cycloalkanes, or heteroleptic complexes. These compounds have different groups attached to the same atoms, which are called stereogenic elements.
- Geometrical isomers can be named using the cis-trans or E-Z nomenclature. The cis-trans nomenclature is based on the relative position of the groups on the same side or opposite sides of the bond or ring. The E-Z nomenclature is based on the priority of the groups according to the Cahn-Ingold-Prelog rules.

Isomerism and Chiral Drugs

- Isomerism is the phenomenon of having two or more compounds with the same molecular formula but different structures or spatial arrangements .
- Isomers can be classified into two main types: constitutional isomers and stereoisomers .
- Constitutional isomers have different connectivity of atoms, while stereoisomers have the same connectivity but different orientation in space .

- Stereoisomers can be further divided into two subtypes: geometric isomers and optical isomers .
- Geometric isomers have different arrangement of atoms around a double bond or a ring, while optical isomers have different arrangement of atoms around a chiral center .
- A chiral center is an atom, usually carbon, that is bonded to four different groups .
- A chiral molecule is a molecule that has at least one chiral center and is not superimposable on its mirror image .
- The mirror images of a chiral molecule are called enantiomers, which are a pair of optical isomers that have opposite configuration at all chiral centers .
- Enantiomers have identical physical and chemical properties, except for their interaction with plane-polarized light and other chiral molecules .
- A mixture of equal amounts of two enantiomers is called a racemic mixture or a racemate, which has no net optical activity .
- Chiral drugs are drugs that contain chiral molecules as active ingredients .
- Chiral drugs can have different biological effects depending on the stereochemistry of the molecule, as different enantiomers can interact differently with chiral receptors, enzymes, and proteins in the body .
- Chiral drugs can be marketed as racemic mixtures, single enantiomers, or mixtures of different proportions of enantiomers .
- The stereoisomeric composition of a chiral drug should be known and the quantitative isomeric composition of the material used in pharmacologic, toxicologic, and clinical studies should be consistent.
- The development of new chiral drugs should consider the potential advantages and disadvantages of using single enantiomers or racemic mixtures, such as efficacy, safety, pharmacokinetics, pharmacodynamics, and cost.
- Some examples of chiral drugs are ibuprofen, salbutamol, thalidomide, and warfarin .

Unit-3: 8

- Unit-3: 8 is a topic that covers the following subtopics:
 - The concept of **functions** and how to define, evaluate, and graph them.
 - The concept of **inverse functions** and how to find and use them.
 - The concept of **exponential functions** and how to model growth and decay phenomena with them.
 - The concept of **logarithmic functions** and how to use their properties and graphs.
 - The concept of **compound interest** and how to calculate the future and present value of an investment.
 - The concept of **continuous compounding** and how to use the natural exponential function and the natural logarithm function.

- The concept of **exponential growth and decay models** and how to use them to analyze real-world situations.
- The concept of **logistic growth models** and how to use them to describe limited growth phenomena.
- Some of the key terms and formulas related to this topic are:
 - A **function** is a rule that assigns to each input value exactly one output value.
 - The **domain** of a function is the set of all possible input values, and the **range** is the set of all possible output values.
 - A function can be represented by an **equation**, a **table**, a **graph**, or a **verbal description**.
 - A function can be **evaluated** by substituting a value for the input variable and simplifying the expression.
 - A function can be **graphed** by plotting ordered pairs that satisfy the equation of the function.
 - A function is **one-to-one** if it passes the **horizontal line test**, which means that no horizontal line intersects the graph of the function more than once.
 - The **inverse** of a function is another function that reverses the input-output relationship of the original function.
 - The inverse of a function can be found by **swapping** the input and output variables and **solving** for the new input variable.
 - The inverse of a function can be **verified** by showing that the composition of the function and its inverse is equal to the identity function, which means that $f(f^{-1}(x)) = x$ and $f^{-1}(f(x)) = x$ for all x in the domain of f and f^{-1} .
 - The inverse of a function can be **graphed** by **reflecting** the graph of the original function over the line $y = x$.
 - An **exponential function** is a function of the form $f(x) = a(b)^x$, where a and b are constants, a is not zero, and b is positive and not equal to one.
 - The **base** of an exponential function is the constant b , and the **initial value** is the constant a .
 - The **growth factor** of an exponential function is the amount by which the function value changes when the input value increases by one unit. It is equal to the base b .
 - The **growth rate** of an exponential function is the percentage by which the function value changes when the input value increases by one unit. It is equal to $(b - 1) * 100\%$.
 - The **decay factor** of an exponential function is the amount by which the function value changes when the input value decreases by one unit. It is equal to the reciprocal of the base b .
 - The **decay rate** of an exponential function is the percentage by which the function value changes when the input value decreases by one unit. It is equal to $(1 - b) * 100\%$.
 - The **end behavior** of an exponential function describes how the

function value approaches infinity or zero as the input value increases or decreases without bound. It depends on the sign of a and the value of b .

- The **asymptote** of an exponential function is the horizontal line that the graph of the function approaches but never touches. It is given by the equation $y = a$.
- A **logarithmic function** is the inverse of an exponential function. It is a function of the form $f(x) = \log_b(x)$, where b is the base of the logarithm and x is the argument of the logarithm.
- The **base** of a logarithmic function is the constant b , and the **argument** is the variable x .
- The **domain** of a logarithmic function is the set of all positive numbers, and the **range** is the set of all real numbers.
- The **logarithmic function** can be **evaluated** by using the **definition** of the logarithm,

Electrochemistry and Batteries: Basic concepts of electrochemistry

- Electrochemistry is a branch of chemistry that deals with the interconversion of chemical energy and electrical energy.
- Electrochemical reactions either absorb or release energy, which can be in the form of electricity.
- Electrochemical cells are devices that use electrochemical reactions to produce or consume electrical energy.
- There are two types of electrochemical cells: galvanic cells and electrolytic cells.
- Galvanic cells, also known as voltaic cells, are electrochemical cells in which spontaneous oxidation-reduction reactions produce electrical energy.
- Electrolytic cells are electrochemical cells in which electrical energy is used to drive non-spontaneous oxidation-reduction reactions.
- Oxidation-reduction reactions, also known as redox reactions, are reactions that involve the transfer of electrons from one substance to another.
- Oxidation is the loss of electrons, while reduction is the gain of electrons.
- In a redox reaction, the substance that loses electrons is called the reducing agent, while the substance that gains electrons is called the oxidizing agent.
- In an electrochemical cell, the oxidation and reduction reactions take place at two different electrodes: the anode and the cathode.
- The anode is the electrode where oxidation occurs, while the cathode is the electrode where reduction occurs.
- The electrodes are connected by an external circuit that allows the flow of electrons from the anode to the cathode.
- The electrodes are also immersed in an electrolyte, which is a solution that contains ions that can conduct electricity.

- The electrolyte allows the flow of ions between the electrodes to maintain electrical neutrality.
- A salt bridge or a porous membrane may be used to separate the electrolytes of different electrodes to prevent mixing.
- The difference in the electrical potential between the anode and the cathode is called the cell potential or the electromotive force (emf) of the cell.
- The cell potential is a measure of the tendency of the cell to produce or consume electrical energy.
- The cell potential can be calculated using the Nernst equation, which relates the cell potential to the concentrations of the reactants and products.
- A battery is a device that consists of one or more electrochemical cells connected in series or parallel to provide a source of electrical energy.
- There are two basic types of batteries: primary and secondary.
- Primary batteries are “single use” and cannot be recharged. They are used for devices that require low power and long shelf life, such as flashlights, toys, and remote controls.
- Secondary batteries are “rechargeable” and can be reused by reversing the electrochemical reactions. They are used for devices that require high power and frequent use, such as laptops, phones, and electric vehicles.
- Some examples of primary batteries are zinc-carbon batteries, alkaline batteries, and lithium batteries.
- Some examples of secondary batteries are lead-acid batteries, nickel-cadmium batteries, and lithium-ion batteries.
- The performance of a battery depends on several factors, such as the type of electrodes, electrolyte, and separator, the capacity, the rate of discharge, the temperature, and the shelf life.

Batteries; Classification and Applications of Primary Cells (Dry Cell) and Secondary Cells

- A battery is a device that converts chemical energy into electrical energy by using one or more electrochemical cells.
- A cell is a basic unit of a battery that consists of two electrodes (anode and cathode) and an electrolyte that allows the flow of ions between the electrodes.
- Batteries can be classified into two main types: primary cells and secondary cells .
- Primary cells are cells that are designed to be used once and discarded, and not recharged with electricity and reused like secondary cells.
- The electrochemical reaction occurring in the primary cell is not reversible, rendering the cell unchargeable.
- Examples of primary cells are dry cells, alkaline cells, lithium cells, zinc-air cells, etc .
- Applications of primary cells include flashlights, toys, remote controls,

clocks, radios, etc .

- Secondary cells are cells that can be recharged with electricity and reused multiple times .
- The electrochemical reaction occurring in the secondary cell is reversible, allowing the cell to be recharged.
- Examples of secondary cells are lead-acid cells, nickel-cadmium cells, nickel-metal hydride cells, lithium-ion cells, etc .
- Applications of secondary cells include automobiles, laptops, mobile phones, cameras, etc .

Lead Acid Battery

A lead acid battery is a type of rechargeable battery that was first invented in 1859 by French physicist Gaston Planté. It is the oldest type of rechargeable battery and is widely used in various applications, such as automotive, grid, and backup power.

Structure and Operation

A lead acid battery consists of two electrodes submerged in an electrolyte of sulfuric acid. The positive electrode is made of grains of metallic lead oxide, while the negative electrode is attached to a grid of metallic lead. The electrodes are separated by a porous separator that prevents short circuits.

When the battery is charged, an electric current is applied to the electrodes, causing the lead oxide to be reduced to lead and the lead to be oxidized to lead sulfate. The sulfuric acid is also consumed and water is produced. The chemical reactions are as follows:

Positive electrode: $\text{PbO}_2 + 4\text{H}^+ + \text{SO}_4^{2-} + 2\text{e}^- \rightarrow \text{PbSO}_4 + 2\text{H}_2\text{O}$

Negative electrode: $\text{Pb} + \text{SO}_4^{2-} \rightarrow \text{PbSO}_4 + 2\text{e}^-$

Overall: $\text{PbO}_2 + \text{Pb} + 2\text{H}_2\text{SO}_4 \rightarrow 2\text{PbSO}_4 + 2\text{H}_2\text{O}$

When the battery is discharged, the reverse reactions occur, producing electricity and restoring the sulfuric acid. The voltage of a single lead acid cell is about 2 V, and multiple cells can be connected in series to increase the voltage.

Types and Applications

Lead acid batteries are classified into two types: flooded and valve-regulated. Flooded batteries have excess electrolyte that covers the electrodes and allows gas to escape during charging. Valve-regulated batteries have a sealed case that prevents gas leakage and uses a valve to release excess pressure. Valve-regulated batteries are further divided into gel and absorbed glass mat (AGM) types, depending on the form of the electrolyte.

Lead acid batteries have relatively low energy density, meaning they are heavy and bulky for the amount of energy they store. However, they have high power density, meaning they can deliver high currents for short periods of time. They are also cheap, durable, and easy to maintain. These characteristics make them suitable for applications that require high power and long lifespan, such as:

- Automotive: Lead acid batteries are used as the main power source for starting, lighting, and ignition (SLI) systems in most vehicles. They can also provide auxiliary power for hybrid and electric vehicles.
- Grid: Lead acid batteries are used for grid energy storage, frequency regulation, and peak shaving. They can help balance the supply and demand of electricity and integrate renewable energy sources.
- Backup: Lead acid batteries are used for uninterruptible power supply (UPS) and emergency lighting systems. They can provide reliable and continuous power in case of power outages or failures.

Advantages and Disadvantages

Some of the advantages of lead acid batteries are:

- Low cost: Lead acid batteries are the cheapest type of rechargeable battery, making them affordable and widely available.
- High power: Lead acid batteries can deliver high currents for short durations, making them suitable for high-power applications.
- Long lifespan: Lead acid batteries can last for several years if properly maintained and operated.
- Easy recycling: Lead acid batteries are highly recyclable, reducing the environmental impact and waste disposal costs.

Some of the disadvantages of lead acid batteries are:

- Low energy: Lead acid batteries have low energy density, meaning they are heavy and bulky for the amount of energy they store. They also have low specific energy, meaning they lose capacity as the discharge rate increases.
- Poor performance: Lead acid batteries have poor performance at low temperatures, high temperatures, and deep discharge cycles. They also suffer from self-discharge, sulfation, and corrosion, which reduce their efficiency and lifespan.
- Environmental hazard: Lead acid batteries contain toxic and corrosive materials, such as lead and sulfuric acid, which pose a risk of leakage, explosion, and pollution. They also require proper handling, storage, and disposal to prevent health and environmental hazards.

Corrosion

Introduction to corrosion

- Corrosion is the deterioration of a material, usually a metal, due to a chemical or electrochemical reaction with its environment.
- Corrosion can cause loss of material, mechanical strength, functionality, and aesthetic value of the metal.
- Corrosion can also pose safety, health, and environmental hazards if not prevented or controlled.
- Corrosion is a natural and inevitable process that can be influenced by many factors, such as the type of metal, the surrounding medium, the temperature, the presence of impurities, the stress, and the time.

Types of corrosion

- There are many types of corrosion, depending on the mechanism, morphology, and environment of the corrosion process. Some common types are:
 - Uniform corrosion: The metal surface corrodes evenly and uniformly, resulting in a thin layer of oxide or corrosion product. This type of corrosion is predictable and easy to measure, but it can cause significant material loss over time.
 - Galvanic corrosion: The metal surface corrodes preferentially at the areas where it is in contact with a more noble metal or a different environment, forming a galvanic cell. This type of corrosion can cause rapid and localized damage, especially in salt water or moist environments.
 - Pitting corrosion: The metal surface corrodes selectively at the areas where there are defects, such as scratches, pores, or inclusions, forming small pits or holes. This type of corrosion can cause perforation and failure of the metal, even if the overall material loss is small.
 - Crevice corrosion: The metal surface corrodes at the areas where there are narrow gaps or crevices between the metal and another material, such as gaskets, washers, or deposits. This type of corrosion can cause severe damage, as the crevice creates a stagnant and acidic environment that accelerates the corrosion process.
 - Intergranular corrosion: The metal surface corrodes along the grain boundaries of the metal, leaving the grains intact. This type of corrosion can cause embrittlement and cracking of the metal, especially in alloys that are susceptible to sensitization, such as stainless steels.
 - Stress corrosion cracking: The metal surface cracks under the combined effect of tensile stress and a corrosive environment. This type

of corrosion can cause catastrophic failure of the metal, as the cracks propagate rapidly and without warning.

- Corrosion fatigue: The metal surface cracks under the combined effect of cyclic stress and a corrosive environment. This type of corrosion can reduce the fatigue life and strength of the metal, as the cracks initiate and grow faster in the presence of corrosion.
- Erosion corrosion: The metal surface corrodes and wears away due to the mechanical action of a fluid or a solid particle that impinges on the metal. This type of corrosion can cause increased roughness and loss of material, especially in high-velocity or turbulent flow conditions.
- Microbiologically influenced corrosion: The metal surface corrodes due to the metabolic activity of microorganisms, such as bacteria, fungi, or algae, that adhere to the metal or form biofilms. This type of corrosion can cause various forms of damage, such as pitting, crevice corrosion, intergranular corrosion, or corrosion under deposits.

Cause of corrosion

- The cause of corrosion is the thermodynamic tendency of metals to revert to their more stable forms, such as oxides, hydroxides, sulfides, or carbonates, in the presence of a suitable oxidizing agent, such as oxygen, water, acids, or salts.
- The corrosion process involves an electrochemical reaction, in which the metal acts as an anode and loses electrons, while the oxidizing agent acts as a cathode and gains electrons. The electrons flow through an external circuit, creating an electric current. The anodic and cathodic reactions can occur on the same or different metal surfaces, depending on the type of corrosion.
- The corrosion rate depends on the kinetics of the electrochemical reaction, which can be influenced by many factors, such as the type and concentration of the oxidizing agent, the pH and temperature of the environment, the presence of inhibitors or catalysts, the surface condition and composition of the metal, and the applied or residual stress on the metal.

Corrosion prevention and control

- Corrosion prevention and control are the methods and techniques used to reduce or eliminate the corrosion damage of metals. Some common methods are:
 - Material selection: Choosing a metal or an alloy that is resistant or immune to the specific corrosive environment, such as stainless steels, aluminum alloys, or titanium alloys.
 - Protective coating: Applying a layer of paint, enamel, plastic, rub-

ber, or metal on the metal surface to isolate it from the corrosive environment,

Prevention and Control of Corrosion

Corrosion is the deterioration of a metal or its properties due to a reaction with its environment. Corrosion can cause loss of material, strength, functionality, and aesthetics of metal structures and devices. Corrosion can also pose safety and environmental hazards if not properly controlled.

Some of the methods to prevent and control corrosion are:

- **Material selection:** Choosing a metal or alloy that is resistant to the specific corrosive environment or has a protective oxide layer. For example, stainless steel, aluminum, and copper are often used for their corrosion resistance.
- **Coating:** Applying a layer of paint, plastic, metal, or ceramic on the metal surface to isolate it from the corrosive environment. For example, steel pipes are often coated with zinc or epoxy to prevent rusting.
- **Cathodic protection:** Connecting the metal to a more easily corroded metal (called a sacrificial anode) or an external power source (called an impressed current) to make the metal a cathode and reduce its corrosion rate. For example, underground pipelines and storage tanks are often protected by cathodic protection systems.
- **Corrosion inhibitors:** Adding chemicals to the corrosive environment to reduce the corrosion rate of the metal. For example, corrosion inhibitors are often added to cooling water systems, oil and gas pipelines, and industrial processes.
- **Design and maintenance:** Designing the metal structure or device to avoid crevices, stress concentrations, galvanic couples, and other factors that can accelerate corrosion. Also, performing regular inspection, cleaning, and repair of the metal to remove corrosion products and prevent further damage. For example, bridges, buildings, and vehicles are often designed and maintained to prevent corrosion.

Issues in Specific Industries (Power Generation, Chemical)

Power Generation

- Power generation is the process of converting energy from various sources (such as fossil fuels, nuclear, renewable, etc.) into electricity that can be transmitted and distributed to consumers.
- Some of the issues in power generation are:

- Environmental impact: Power generation can cause air pollution, greenhouse gas emissions, water consumption, waste generation, and land use, which can affect the climate, biodiversity, and human health. Some power sources, such as coal and nuclear, have higher environmental impacts than others, such as wind and solar.
- Reliability and security: Power generation needs to be reliable and secure to meet the demand and avoid blackouts, brownouts, or cyber-attacks. Some power sources, such as solar and wind, are intermittent and depend on weather conditions, which can affect their availability and stability. Other power sources, such as hydro and nuclear, are vulnerable to natural disasters, such as floods and earthquakes, which can damage their infrastructure and cause safety risks.
- Cost and efficiency: Power generation needs to be cost-effective and efficient to provide affordable and accessible electricity to consumers and industries. Some power sources, such as fossil fuels and nuclear, have high capital and operational costs, which can increase the electricity prices and burden the consumers. Other power sources, such as renewable, have low operational costs, but may require subsidies, incentives, or grid integration to be competitive and viable.
- Innovation and regulation: Power generation needs to be innovative and regulated to adapt to the changing needs and preferences of the consumers and the society. Some power sources, such as renewable, have high potential for innovation and development, which can improve their performance and reduce their costs. Other power sources, such as fossil fuels and nuclear, have low potential for innovation and development, but may face regulatory and social barriers, such as public opposition, legal challenges, or political instability.

Chemical

- Chemical is the branch of science and engineering that deals with the composition, structure, properties, and reactions of matter, especially of atomic and molecular systems.
- Some of the issues in chemical are:
 - Safety and health: Chemical can pose safety and health risks to the workers, consumers, and communities, who may be exposed to hazardous substances, such as toxic, flammable, corrosive, or explosive materials. Chemical accidents, such as spills, leaks, fires, or explosions, can cause injuries, fatalities, or environmental damage. Chemical safety and health require proper handling, storage, transportation, and disposal of chemicals, as well as personal protective equipment, emergency response, and risk assessment.
 - Sustainability and ethics: Chemical can have negative impacts on the environment and the society, such as resource depletion, pollution, waste generation, and social injustice. Chemical sustainability and

ethics require the development and implementation of green chemistry, which is the design of chemical products and processes that reduce or eliminate the use and generation of hazardous substances, and the consideration of the social and environmental implications of chemical activities, such as human rights, animal welfare, and environmental justice.

- Innovation and competitiveness: Chemical needs to be innovative and competitive to meet the demand and expectations of the consumers and the industries, who may require new or improved chemical products and processes, such as biotechnology, nanotechnology, or materials science. Chemical innovation and competitiveness require the investment and support of research and development, education and training, intellectual property rights, and market access.

Processing Industry, Oil & Gas Industry and Pulp & Paper Industry

- Processing industry is a broad term that covers various sectors that transform raw materials into finished products, such as food, beverages, textiles, chemicals, metals, plastics, etc.
- Oil & gas industry is a sector that explores, extracts, refines, transports, and markets petroleum and natural gas products, such as gasoline, diesel, jet fuel, lubricants, asphalt, etc.
- Pulp & paper industry is a sector that produces pulp from wood or other fibrous materials, and manufactures paper and paperboard products, such as newsprint, printing paper, packaging paper, tissue paper, etc.

Processing Industry

- The processing industry is a major contributor to the global economy, as it adds value to the raw materials and creates a variety of products for different markets and consumers.
- The processing industry is also a significant user of energy, water, and other resources, and generates various environmental impacts, such as greenhouse gas emissions, waste, and pollution.
- The processing industry faces many challenges and opportunities, such as increasing demand, changing consumer preferences, technological innovation, digital transformation, sustainability, and competitiveness.

Oil & Gas Industry

- The oil & gas industry is a vital source of energy for the world, as it supplies about 55% of the global primary energy demand, according to the International Energy Agency (IEA).

- The oil & gas industry is also a major employer, investor, and taxpayer, as it supports millions of jobs, contributes to public revenues, and funds infrastructure and social development projects.
- The oil & gas industry faces many challenges and opportunities, such as declining reserves, volatile prices, geopolitical risks, environmental regulations, energy transition, and innovation.

Pulp & Paper Industry

- The pulp & paper industry is a key provider of materials for communication, education, hygiene, and packaging, as it produces about 420 million tonnes of paper and paperboard annually, according to the Food and Agriculture Organization (FAO).
- The pulp & paper industry is also a major user of wood, water, and energy, and generates various environmental impacts, such as deforestation, greenhouse gas emissions, waste, and pollution.
- The pulp & paper industry faces many challenges and opportunities, such as declining demand, digitalization, recycling, innovation, and sustainability.

Chemistry of Engineering Materials

Chemistry of engineering materials is the study of the chemical composition, structure, and properties of materials that are used for engineering applications. Engineering materials include metals, ceramics, polymers, composites, and nanomaterials.

Some of the topics covered in chemistry of engineering materials are:

- Chemical composition of materials: This refers to the elements and compounds that make up a material and their relative proportions. Chemical composition affects the physical and mechanical properties of materials, such as strength, hardness, ductility, brittleness, corrosion resistance, and weldability.
- Acidity or alkalinity of materials: This refers to the pH value of a material, which measures the concentration of hydrogen ions in a solution. pH value ranges from 0 to 14, with 7 being neutral, lower than 7 being acidic, and higher than 7 being alkaline. Acidity or alkalinity affects the chemical reactivity and stability of materials, as well as their compatibility with other materials.
- Phase diagrams of materials: This refers to the graphical representation of the different phases or states of matter (solid, liquid, gas) that a material can exist in under different conditions of temperature and pressure. Phase diagrams show the equilibrium relationships between the phases and the phase transitions that occur when the conditions change. Phase diagrams are useful for understanding the thermodynamics and kinetics of materials,

as well as designing and optimizing processes such as melting, casting, annealing, and quenching.

- **Crystal structure of materials:** This refers to the arrangement of atoms or molecules in a material that forms a repeating pattern in three dimensions. Crystal structure determines the symmetry and shape of a material, as well as its physical and mechanical properties, such as elasticity, thermal expansion, electrical conductivity, and optical properties. Crystal structure can be described by the unit cell, which is the smallest repeating unit of the pattern, and the lattice, which is the infinite array of unit cells.
- **Defects and imperfections in materials:** This refers to the deviations from the ideal crystal structure or composition of a material that are caused by various factors, such as impurities, vacancies, interstitials, dislocations, grain boundaries, and cracks. Defects and imperfections affect the properties and performance of materials, as they can introduce stress, strain, distortion, and diffusion in the material. Some defects can be beneficial, such as doping in semiconductors, while some can be detrimental, such as corrosion in metals.
- **Energy materials:** This refers to the materials that are used for energy conversion, storage, and transmission, such as solar cells, batteries, catalysts, and superconducting magnets. Energy materials are designed to have high efficiency, durability, and sustainability, as well as low cost and environmental impact. Energy materials can be classified by their use in conventional, advanced, and possible future energy systems .

Cement; Constituents, manufacturing, hardening and setting, deterioration of cement, Plaster

- Cement is a binder material that sets and hardens by chemical reactions with water and other ingredients.
- The main constituents of cement are calcium silicates, aluminates, and sulfates, which are derived from limestone, clay, and gypsum.
- The manufacturing of cement involves four main stages: raw material preparation, clinker production, clinker grinding, and cement packing.
 - Raw material preparation involves crushing, grinding, and blending of limestone, clay, and other additives to obtain a homogeneous mixture.
 - Clinker production involves heating the raw material mixture in a rotary kiln at high temperatures (1400-1500°C) to form nodules of clinker, which are mainly calcium silicates and aluminates.
 - Clinker grinding involves cooling and grinding the clinker with gypsum and other additives to obtain the final product of cement.
 - Cement packing involves filling and sealing the cement in bags or containers for storage and transportation.
- Hardening and setting of cement are two different processes that occur

after the cement is mixed with water and other ingredients.

- Setting is the initial stiffening of the cement paste, which occurs within a few hours after mixing. Setting is influenced by the type and amount of cement, water, and additives, as well as the temperature and humidity of the environment.
- Hardening is the gradual increase in strength and durability of the cement paste, which occurs over days, weeks, or months after setting. Hardening is influenced by the hydration and crystallization of the cement compounds, as well as the curing conditions and external factors such as moisture, temperature, and load.
- Deterioration of cement is the loss of quality and performance of the cement paste or concrete due to various factors, such as:
 - Chemical attack: the reaction of the cement paste or concrete with aggressive substances, such as acids, sulfates, chlorides, carbon dioxide, etc., which can cause corrosion, leaching, expansion, cracking, or weakening of the material.
 - Physical damage: the exposure of the cement paste or concrete to mechanical stress, such as abrasion, erosion, impact, fatigue, etc., which can cause wear, tear, or fracture of the material.
 - Thermal effects: the exposure of the cement paste or concrete to extreme or fluctuating temperatures, such as fire, frost, or thermal shock, which can cause expansion, contraction, spalling, or cracking of the material.
- Plaster is a mixture of cement, sand, water, and sometimes other additives, such as lime, fibers, or pigments, that is applied to a surface to form a smooth or decorative coating.
 - Plaster can be used for various purposes, such as protection, insulation, decoration, or repair of walls, ceilings, floors, or other structures.
 - Plaster can be classified into different types, such as gypsum plaster, lime plaster, cement plaster, etc., depending on the type of binder and additives used.
 - Plaster can be applied in different ways, such as by hand, by machine, or by spray, depending on the consistency and properties of the mixture and the desired finish.

Paris

Paris is the capital and most populous city of France, with an estimated population of 2,165,423 residents in 2019 in an area of more than 105 km² (41 sq mi), making it the fourth-most populated city in the European Union as well as the 30th most densely populated city in the world in 2022. Paris is a global center for art, fashion, gastronomy, and culture, and is often regarded as the most romantic city in the world.

Some of the main points to know about Paris are:

- Paris is situated in the north-central part of France, along the Seine River, which divides the city into two parts: the Right Bank and the Left Bank.
- Paris has a rich and turbulent history, dating back to the 3rd century BCE, when it was founded by a Celtic tribe called the Parisii. Paris was the capital of the Frankish kingdom, the French monarchy, the First French Republic, the First French Empire, and the Third French Republic, and witnessed many important events such as the French Revolution, the Reign of Terror, the Napoleonic Wars, the Paris Commune, the World Wars, and the May 1968 protests.
- Paris is home to many iconic landmarks, monuments, museums, and cultural sites, such as the Eiffel Tower, the Arc de Triomphe, the Notre-Dame Cathedral, the Louvre Museum, the Musée d'Orsay, the Sacré-Cœur Basilica, the Palace of Versailles, and the Champs-Élysées.
- Paris is also known for its distinctive architecture, which reflects its historical and artistic influences, such as the Gothic, Renaissance, Baroque, Neo-classical, Haussmannian, Art Nouveau, Art Deco, Modernist, and Post-modernist styles.
- Paris is a major hub for fashion, design, and luxury, hosting many prestigious events such as the Paris Fashion Week, the Paris Haute Couture Week, and the Paris Biennale. Paris is also the headquarters of many renowned fashion houses, such as Chanel, Dior, Louis Vuitton, Hermès, and Yves Saint Laurent.
- Paris is famous for its culinary scene, which offers a variety of dishes, from traditional French cuisine to international specialties. Some of the most popular foods to try in Paris are macarons, croissants, frog legs, steak and frites, eclairs, hot chocolate, French onion soup, and madeleines. Paris is also a great place to buy souvenirs, such as cheese, dried sausage, French chocolate, French macarons, wine, a marinière, a beret, or French candy.
- Paris has a vibrant and diverse cultural life, with many festivals, concerts, theaters, operas, cinemas, art galleries, libraries, and bookstores. Paris is also the birthplace or the residence of many famous artists, writers, musicians, philosophers, and scientists, such as Victor Hugo, Voltaire, Émile Zola, Marcel Proust, Charles Baudelaire, Claude Monet, Edgar Degas, Auguste Rodin, Pierre and Marie Curie, Édith Piaf, Coco Chanel, Jean-Paul Sartre, Simone de Beauvoir, and many more.
- Paris is a popular tourist destination, attracting more than 30 million visitors every year. There are many itineraries to explore Paris, depending on the interests and preferences of the travelers. Some of the most common itineraries are:
 - Rejuvenate with your family in Paris: This itinerary covers some of the most family-friendly attractions in Paris, such as the Eiffel Tower,

the Louvre Museum, the Musée d'Orsay, the Notre-Dame Cathedral, the Arc de Triomphe, and the Sacré-Cœur Basilica.

- Bask in the iconic romantic vibe of Paris: This itinerary focuses on the most romantic places in Paris, such as the Cathédrale Notre-Dame de Paris, the Louvre Museum, the Musée d'Orsay, the Jardin des Tuileries, the Champs-Élysées, and the Arc de Triomphe.
- Get amazed by the historical monuments in Paris and Rome: This itinerary combines the historical and cultural highlights of Paris and Rome, such as the Eiffel

Unit-4: 8

- This unit covers the topic of **linear programming**, which is a method of finding the optimal solution to a problem that involves multiple constraints and objectives.
- Linear programming can be used to model and solve various real-world problems, such as production planning, resource allocation, transportation, scheduling, etc.
- The basic idea of linear programming is to **maximize** or **minimize** a linear function of decision variables, subject to a set of linear equations or inequalities that represent the constraints of the problem.
- A linear function is a function that can be written as a sum of constant multiples of the decision variables, such as $f(x,y) = 3x + 5y$.
- A linear equation is an equation that can be written as a sum of constant multiples of the decision variables equal to a constant, such as $2x + y = 10$.
- A linear inequality is an inequality that can be written as a sum of constant multiples of the decision variables less than or equal to a constant, such as $x + 2y \leq 8$.
- The decision variables are the unknown quantities that need to be determined by the linear programming method, such as the amount of products to produce, the number of workers to hire, the routes to take, etc.
- The constraints are the limitations or restrictions that the decision variables must satisfy, such as the availability of resources, the demand of customers, the capacity of facilities, etc.
- The objective function is the linear function that represents the goal of the problem, such as the profit to maximize, the cost to minimize, the efficiency to optimize, etc.
- A feasible solution is a set of values for the decision variables that satisfies all the constraints of the problem.
- An optimal solution is a feasible solution that gives the best possible value for the objective function.
- A feasible region is the set of all feasible solutions, which can be represented graphically by a polygonal region on a coordinate plane.
- A corner point is a point on the boundary of the feasible region that has at least two adjacent sides that are not parallel.

- An optimal solution can always be found at a corner point of the feasible region, if it exists.
- A graphical method can be used to solve linear programming problems with two decision variables, by plotting the feasible region and the objective function on the same graph, and finding the corner point that gives the highest or lowest value for the objective function, depending on whether the problem is a maximization or a minimization problem.
- A simplex method can be used to solve linear programming problems with more than two decision variables, by using a systematic algorithm that moves from one corner point to another, until the optimal solution is reached or the problem is proven to be unbounded or infeasible.

Water Technology: Sources and impurities of water, Hardness of water, Boiler troubles

Sources and impurities of water

- Water is a vital resource for human and industrial activities. It is obtained from various sources, such as surface water (rivers, lakes, ponds, etc.), groundwater (wells, springs, etc.), rainwater, seawater, etc.
- Water from different sources may contain various impurities, such as dissolved solids (salts, minerals, metals, etc.), dissolved gases (oxygen, carbon dioxide, hydrogen sulfide, etc.), suspended solids (sand, clay, silt, etc.), colloidal particles (organic matter, microorganisms, etc.), and dissolved organic compounds (pesticides, herbicides, etc.).
- The impurities in water may affect its quality, suitability, and safety for various purposes, such as drinking, domestic use, industrial use, agricultural use, etc. Therefore, water quality analysis and treatment are essential to ensure the desired water quality standards.

Hardness of water

- Hardness of water is a measure of the concentration of dissolved calcium and magnesium salts in water. These salts react with soap to form insoluble precipitates, called scum or curd, which reduce the lathering and cleaning ability of soap. Hard water also causes scaling or deposition of calcium and magnesium salts on the surfaces of pipes, boilers, heat exchangers, etc., which reduces their efficiency and lifespan.
- Hardness of water is classified into two types: temporary hardness and permanent hardness. Temporary hardness is caused by the presence of bicarbonates of calcium and magnesium, which can be removed by boiling the water or adding lime. Permanent hardness is caused by the presence of sulfates, chlorides, and nitrates of calcium and magnesium, which cannot be removed by boiling or lime treatment, but require other methods, such as ion exchange, reverse osmosis, etc.

- Hardness of water is expressed in terms of equivalent calcium carbonate (CaCO_3) concentration in milligrams per liter (mg/L) or parts per million (ppm). The degree of hardness of water is generally categorized as follows:

Degree of hardness	Hardness range (mg/L or ppm)
Soft	0 - 75
Moderately hard	75 - 150
Hard	150 - 300
Very hard	> 300

Boiler troubles

- Boiler troubles are the problems or defects that occur in boilers due to the presence of impurities in the feed water or the improper operation or maintenance of the boiler system. Some of the common boiler troubles are:
 - Scale formation: Scale is the hard and adherent coating of calcium and magnesium salts, such as calcium sulfate, magnesium hydroxide, and calcium bicarbonate, that forms on the inner walls of the boiler tubes and vessels. Scale reduces the heat transfer efficiency and increases the fuel consumption and operating cost of the boiler. Scale also increases the risk of overheating and bursting of the boiler tubes. Scale can be prevented or removed by using soft water, adding scale inhibitors, or mechanical cleaning.
 - Sludge formation: Sludge is the soft and loose deposit of insoluble impurities, such as silica, iron oxide, and organic matter, that settles at the bottom or low-lying areas of the boiler. Sludge reduces the water circulation and heat transfer efficiency of the boiler. Sludge also causes corrosion and pitting of the boiler metal. Sludge can be prevented or removed by using clarified water, adding sludge conditioners, or blowdown operation.
 - Corrosion: Corrosion is the deterioration or damage of the boiler metal due to the chemical or electrochemical reaction with the water or the dissolved gases, such as oxygen and carbon dioxide, in the water. Corrosion reduces the strength and durability of the boiler metal and causes leaks and failures of the boiler. Corrosion can be prevented or controlled by using deaerated water, adding corrosion inhibitors, or cathodic protection.
 - Priming and foaming: Priming is the phenomenon of carrying over of water droplets along with the steam from the boiler. Foaming is the formation of stable bubbles on the surface of the water in the boiler. Both priming and foaming are caused by the presence of high

concentration of dissolved solids, suspended solids, or organic matter in the water. Priming and foaming reduce the quality and quantity of the steam and cause wetness, erosion, and corrosion of the steam pipes and turbines. Priming and foaming can be prevented or

Techniques for Water Softening

Water softening is the process of removing or reducing the hardness of water, which is caused by the presence of dissolved calcium and magnesium ions. Hard water can cause problems such as scaling, corrosion, soap scum formation, and reduced efficiency of water heaters and appliances. There are several techniques for water softening, each with its own advantages and disadvantages. Some of the most common techniques are:

- **Lime-Soda Softening:** This is a chemical method that involves adding lime (calcium hydroxide) and soda ash (sodium carbonate) to the hard water. The lime reacts with the calcium and magnesium ions to form insoluble precipitates of calcium carbonate and magnesium hydroxide, which can be removed by filtration or sedimentation. The soda ash reacts with any remaining calcium ions to form soluble sodium carbonate, which does not contribute to hardness. This method can reduce the hardness of water to less than 1 ppm, but it requires large amounts of chemicals, produces a lot of sludge, and may increase the alkalinity and pH of the water .
- **Zeolite Softening:** This is a physical method that uses a natural or synthetic resin, called zeolite, that has a high affinity for calcium and magnesium ions. The hard water is passed through a column of zeolite, which exchanges its sodium ions for the calcium and magnesium ions in the water. The zeolite is then regenerated by flushing it with a concentrated salt solution, which restores its sodium ions and removes the calcium and magnesium ions. This method can reduce the hardness of water to less than 5 ppm, but it requires periodic regeneration, produces a salty waste stream, and may increase the sodium content of the water .
- **Ion Exchange Softening:** This is a similar method to zeolite softening, but it uses a synthetic resin that can exchange any cation (positively charged ion) for another. The hard water is passed through a column of cation exchange resin, which exchanges its hydrogen ions for the calcium and magnesium ions in the water. The resin is then regenerated by flushing it with a strong acid, such as hydrochloric acid or sulfuric acid, which restores its hydrogen ions and removes the calcium and magnesium ions. This method can reduce the hardness of water to less than 1 ppm, but it requires periodic regeneration, produces an acidic waste stream, and may lower the pH of the water .
- **Reverse Osmosis Softening:** This is a membrane-based method that uses a semi-permeable membrane that allows water molecules to pass

through, but blocks most dissolved salts and impurities. The hard water is subjected to a high pressure, which forces it through the membrane, leaving behind a concentrated solution of calcium and magnesium ions, called brine. The water that passes through the membrane, called permeate, is soft and pure. This method can reduce the hardness of water to less than 1 ppm, but it requires a lot of energy, produces a lot of brine, and may remove some beneficial minerals from the water .

Process of Determination of Hardness and Alkalinity

Alkalinity

- Alkalinity is the measure of the ability of water to neutralize acids.
- Alkalinity is mainly due to the presence of bicarbonates, carbonates, and hydroxides of calcium, magnesium, and other metals in water.
- Alkalinity is expressed as milligrams per liter of equivalent calcium carbonate (mg/L CaCO₃).
- Alkalinity can be measured by titrating a water sample with a standard acid solution, such as sulfuric acid (H₂SO₄), using a pH meter or an indicator to detect the endpoint.
- The endpoint of the titration depends on the type and amount of alkaline substances in water.
- The most common endpoints are pH 4.5, pH 8.3, and pH 10.0, corresponding to total, carbonate, and bicarbonate alkalinity, respectively.
- The titration curve of alkalinity shows the change of pH with the addition of acid.
- The inflection points of the curve indicate the equivalence points of the titration, where the moles of acid added are equal to the moles of alkaline substances neutralized.
- The alkalinity can be calculated from the volume and concentration of the acid used and the volume of the water sample.
- The formula for alkalinity calculation is:
$$\text{Alkalinity (mg/L CaCO}_3\text{)} = (\text{Volume of acid (mL)} \times \text{Normality of acid (eq/L)} \times 50,000) / \text{Volume of water sample (mL)}$$
- Some tips for alkalinity measurement are:
 - Use fresh and standardized acid solution.
 - Rinse the burette and the pH electrode with acid and water before use.

- Stir the water sample constantly during the titration.
- Record the initial and final readings of the burette and the pH meter.
- Repeat the titration until consistent results are obtained.

Hardness

- Hardness is the measure of the concentration of calcium and magnesium ions in water.
- Hardness is mainly due to the dissolution of limestone, dolomite, gypsum, and other minerals in water.
- Hardness is expressed as milligrams per liter of equivalent calcium carbonate (mg/L CaCO₃).
- Hardness can be measured by titrating a water sample with a standard complexing agent, such as ethylenediaminetetraacetic acid (EDTA), using an indicator that changes color when all the calcium and magnesium ions are complexed.
- The indicator used for hardness titration is usually eriochrome black T (EBT), which forms a red complex with free calcium and magnesium ions, and turns blue when they are bound to EDTA.
- The endpoint of the titration is the color change from red to blue.
- The titration curve of hardness shows the change of EBT concentration with the addition of EDTA.
- The equivalence point of the curve is where the EBT concentration is zero, and the moles of EDTA added are equal to the moles of calcium and magnesium ions in water.
- The hardness can be calculated from the volume and concentration of the EDTA used and the volume of the water sample.
- The formula for hardness calculation is:

$$\text{Hardness (mg/L CaCO}_3\text{)} = (\text{Volume of EDTA (mL)} \times \text{Normality of EDTA (eq/L)} \times 50,000) / \text{Volume of water sample (mL)}$$

- Some tips for hardness measurement are:
 - Use fresh and standardized EDTA solution.
 - Rinse the burette and the flask with EDTA and water before use.
 - Add a buffer solution to adjust the pH of the water sample to 10.
 - Add a few drops of EBT indicator to the water sample and swirl to mix.
 - Titrate the water sample with EDTA until the color changes from red to blue.
 - Record the initial and final readings of the burette.
 - Repeat the titration until consistent results are obtained.

Numerical Problems

- Here are some examples of numerical problems on alkalinity and hardness measurement:
 - Example 1: A water sample of 100 mL was titrated with 0.02 N H₂SO₄ to pH 8.3 and 4.5, using 12.5 mL and 18.7 mL of acid, respectively. Calculate the total, carbonate, and bicarbonate alkalinity of the water sample.

Solution:

$$\text{Total alkalinity} = (18.7 \text{ mL} \times 0.02 \text{ eq/L} \times 50,000 \text{ mg})$$

Fuels and Combustion: Definition, Classification, Characteristics of a good fuel, Calorific

- **Fuel** is a material that produces heat and energy on combustion.
- **Combustion** is a chemical reaction in which a fuel combines with oxygen and releases heat and light.
- **Classification of fuels:** Fuels can be classified into three main types based on their physical state, origin and composition.
 - **Solid fuels:** These are fuels that are solid at room temperature, such as coal, wood, charcoal, etc. They have a high carbon content and low hydrogen content. They are usually cheap and abundant, but they produce a lot of smoke and ash on combustion.
 - **Liquid fuels:** These are fuels that are liquid at room temperature, such as petrol, diesel, kerosene, etc. They have a high hydrogen content and low carbon content. They are usually more expensive and less abundant than solid fuels, but they produce less smoke and ash on combustion. They are also easier to transport and store.
 - **Gaseous fuels:** These are fuels that are gaseous at room temperature, such as natural gas, biogas, hydrogen, etc. They have a high hydrogen content and low carbon content. They are usually the most expensive and least abundant of the three types of fuels, but they produce the least amount of smoke and ash on combustion. They are also the most efficient and cleanest of the three types of fuels.
- **Characteristics of a good fuel:** A good fuel is one that meets the following criteria :
 - It should be cheap, easily available and readily combustible.
 - It should have a high calorific value, which is the quantity of heat produced by the combustion of a unit mass of fuel.
 - It should not produce harmful gases or residues that pollute the environment, such as carbon monoxide, sulphur dioxide, nitrogen oxides, etc.
 - It should be dry and have less moisture content, as dry fuel increases

- its calorific value and reduces the loss of heat.
- It should have a low ignition temperature, which is the minimum temperature at which a fuel catches fire spontaneously.
- It should have a moderate rate of combustion, which is the speed at which a fuel burns. A fuel that burns too fast or too slow may cause problems in controlling the flame and the heat output.
- It should have a high flame temperature, which is the maximum temperature attained by the flame during combustion. A high flame temperature indicates a high efficiency of the fuel.
- It should have a low storage and transportation cost, which depends on the physical state, volatility and stability of the fuel.
- **Calorific value:** It is the quantity of heat produced by the combustion of a unit mass of fuel. It is usually expressed in kilojoules per kilogram (kJ/kg) or megajoules per kilogram (MJ/kg). The calorific value of a fuel depends on its composition, moisture content, purity and oxygen content .
- **Higher calorific value (HCV):** It is the amount of heat produced by the complete combustion of a unit mass of fuel when the water vapour produced is condensed and the heat released is recovered. It is also known as the gross calorific value (GCV) or the enthalpy of combustion.
- **Lower calorific value (LCV):** It is the amount of heat produced by the complete combustion of a unit mass of fuel when the water vapour produced is not condensed and the heat released is not recovered. It is also known as the net calorific value (NCV) or the internal energy of combustion.
- The difference between the HCV and the LCV of a fuel is equal to the latent heat of vaporization of water multiplied by the mass of water vapour produced per unit mass of fuel. The LCV is always lower than the HCV. The LCV is more useful for practical purposes, as it represents the actual amount of heat available from the fuel.

Values, Gross & Net calorific value, Determination of calorific value by Bomb Calorimeter

- Values are the principles or standards of behavior that guide one's actions and judgments. They are often based on one's beliefs, preferences, culture, or morality. Values can vary from person to person, group to group, or society to society. Some examples of values are honesty, loyalty, freedom, equality, justice, etc.
- Calorific value is the amount of heat or energy produced by a fuel during complete combustion. It is a variable that represents the amount of heat or energy released and is measured in Gross Calorific Value (GCV) or Net Calorific Value (NCV) .

- Gross calorific value (GCV) is the total amount of heat or energy released when a unit mass or volume of fuel is burned in the presence of oxygen. It includes the latent heat of vaporization of water vapors produced during combustion. GCV is also known as higher heating value (HHV) or higher calorific value (HCV) .
- Net calorific value (NCV) is the amount of heat or energy released when a unit mass or volume of fuel is burned in the presence of oxygen, minus the latent heat of vaporization of water vapors produced during combustion. NCV is also known as lower heating value (LHV) or lower calorific value (LCV) .
- The formula for calorific value is: $\text{GCV} - \text{latent heat of water vapors} = \text{NCV}$.
- A bomb calorimeter is an instrument used to determine the calorific value of a fuel. It consists of a strong metal container called a bomb, which is filled with oxygen and a sample of fuel. The bomb is immersed in a known mass of water in an insulated vessel. The fuel is ignited by an electric spark and burns completely in the oxygen. The heat released by the combustion is transferred to the water and the temperature rise of the water is measured. The calorific value of the fuel can be calculated by using the following formula: $\text{Calorific value} = (\text{heat capacity of water} \times \text{mass of water} \times \text{temperature rise of water}) / \text{mass of fuel}$.

Theoretical calculation of calorific value by Dulong's method, Ranking of Coal, Analysis of coal

Theoretical calculation of calorific value by Dulong's method

- Calorific value (CV) is the amount of heat released by the complete combustion of a unit mass of a fuel.
- Dulong's method is a formula that estimates the CV of a fuel from its ultimate analysis, which is the percentage of carbon (C), hydrogen (H), oxygen (O), nitrogen (N) and sulfur (S) in the fuel.
- Dulong's method is based on the assumption that the heat of combustion of an organic compound is nearly equal to the heats of combustion of the elements in it, multiplied by their percentage content in the compound.
- Dulong's formula for CV is:

$$\text{CV} = 337C + 1442(H - O/8) + 93S$$

where C, H, O and S are the percentage of carbon, hydrogen, oxygen and sulfur in the fuel, respectively, and CV is in kJ/kg.

- Dulong's formula has some limitations, such as:

- It does not account for the moisture and ash content of the fuel, which affect the CV.
- It does not account for the variation in the heats of combustion of different forms of carbon, such as graphite, coal and coke.
- It does not account for the variation in the heats of combustion of different forms of hydrogen, such as gas, liquid and solid.
- It does not account for the variation in the heats of combustion of different forms of sulfur, such as pyrite, sulfate and sulfide.
- Therefore, Dulong's formula may not be accurate for all types of fuels, especially those with high moisture, ash, nitrogen or oxygen content.
- Some modifications of Dulong's formula have been proposed to improve its accuracy, such as:
 - Adding a correction factor for the moisture content of the fuel, such as:

$$CV = 337C + 1442(H - O/8) + 93S - 18.6M$$
 where M is the percentage of moisture in the fuel.
 - Adding a correction factor for the ash content of the fuel, such as:

$$CV = 337C + 1442(H - O/8) + 93S - 18.6M - 0.212A$$
 where A is the percentage of ash in the fuel.
 - Adding a correction factor for the nitrogen content of the fuel, such as:

$$CV = 337C + 1442(H - O/8) + 93S - 18.6M - 0.212A - 23.2N$$
 where N is the percentage of nitrogen in the fuel.
 - Using different coefficients for different forms of carbon, hydrogen and sulfur, such as:

$$CV = 337C_g + 326C_c + 314C_k + 1442(H - O/8) + 93S_s + 71S_p + 25S_o$$
 where C_g, C_c and C_k are the percentage of graphite, coal and coke in the fuel, respectively, and S_s, S_p and S_o are the percentage of sulfide, sulfate and sulfur in the fuel, respectively.

Ranking of Coal

- Coal is a combustible sedimentary rock that is formed from the accumulation and compression of plant matter over millions of years.
- Coal is classified into different ranks based on its degree of metamorphism, which is the process of physical and chemical changes that occur due to heat and pressure.

- The degree of metamorphism affects the properties of coal, such as its carbon content, volatile matter, moisture, ash, calorific value, hardness, coking ability and combustion characteristics.
- The main ranks of coal are:
 - Peat: The lowest rank of coal, which is not technically a coal, but a partially decomposed plant matter that has a high moisture content (up to 75%), low carbon content (up to 60%), low calorific value (about 10 MJ/kg) and high ash content (up to 30%). Peat is used as a fuel for domestic heating and cooking, and as a soil conditioner and fertilizer.
 - Lignite: The lowest rank of true coal, which is brown or black in color, soft and crumbly, and has a high moisture content (up to 45%), low carbon content (up to 70%), low calorific value (about 15 MJ/kg) and high ash content (up to 20%). Lignite is used as

Proximate and Ultimate Analysis of Coal

Proximate Analysis

- Proximate analysis is the technique used to analyze the **moisture content, ash content, volatile matter and fixed carbon** of coal.
- It is reported as **dry weight**.
- It is used on coals during **exploration, mining and preparation**.
- It is performed in a **proximate analyzer**, which is an automated instrument.
- The steps involved in proximate analysis are:
 - **Moisture content**: The coal sample is dried in an oven at 105°C for 1 hour and the weight loss is recorded as moisture content.
 - **Ash content**: The dried coal sample is burned in a furnace at 750°C for 3 hours and the remaining residue is weighed as ash content.
 - **Volatile matter**: The dried coal sample is heated in a crucible with a lid at 950°C for 7 minutes and the weight loss is recorded as volatile matter.
 - **Fixed carbon**: The fixed carbon content is calculated by subtracting the moisture, ash and volatile matter from 100.

Ultimate Analysis

- Ultimate analysis is the technique used to analyze the **chemical composition** of coal, such as **carbon, hydrogen, nitrogen, sulfur and oxygen**.
- It is reported as **mass fraction**.
- It is used to determine the **calorific value, combustion efficiency and environmental impact** of coal.

- It is performed in an **ultimate analyzer**, which measures the weight percent of each element by combustion in oxygen.
- The steps involved in ultimate analysis are:
 - **Carbon and hydrogen:** A coal sample is burned in oxygen in a bomb calorimeter and the resulting gases are collected and analyzed by gas chromatography or infrared spectroscopy.
 - **Nitrogen:** A coal sample is burned in oxygen in a bomb calorimeter and the resulting gases are passed through a copper oxide tube to convert nitrogen oxides to nitrogen gas, which is measured by thermal conductivity or chemiluminescence.
 - **Sulfur:** A coal sample is burned in oxygen in a bomb calorimeter and the resulting gases are passed through a magnesium perchlorate tube to absorb water vapor, then through a barium chloride solution to precipitate barium sulfate, which is weighed as sulfur content.
 - **Oxygen:** The oxygen content is calculated by subtracting the carbon, hydrogen, nitrogen and sulfur from 100.

Numerical Problems

- Example 1: A coal sample has the following proximate analysis: moisture 5%, ash 10%, volatile matter 35%, fixed carbon 50%. Calculate the dry and dry ash free basis of the coal sample.
 - Solution:
 - * Dry basis = $(100 - \text{moisture}) / 100 = 0.95$
 - * Dry ash free basis = $(100 - \text{moisture} - \text{ash}) / 100 = 0.85$
- Example 2: A coal sample has the following ultimate analysis: carbon 75%, hydrogen 5%, nitrogen 1%, sulfur 2%, oxygen 17%. Calculate the calorific value of the coal sample in kJ/kg, assuming complete combustion and using the following heats of formation: $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$, -393.5 kJ/mol; $\text{H}_2 + 0.5 \text{O}_2 \rightarrow \text{H}_2\text{O}$, -285.8 kJ/mol; $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$, -296.8 kJ/mol.
 - Solution:
 - * Calorific value = $(75/12) \times (-393.5) + (5/2) \times (-285.8) + (2/32) \times (-296.8) = -32188.8 \text{ kJ/kg}$

Chemistry of Biogas

- Biogas is a mixture of gases produced by the **anaerobic digestion** of organic matter, such as animal manure, sewage, crop residues, etc.
- The main components of biogas are **methane (CH₄) and carbon dioxide (CO₂)**, with minor amounts of hydrogen sulfide (H₂S), ammonia (NH₃), hydrogen (H₂), nitrogen (N₂), etc.
- The typical composition of biogas is: CH₄ 50-70%, CO₂ 30-50%, H₂S 0

Production from organic waste materials and their environmental impact on society

- Organic waste materials are any biodegradable substances that come from plants or animals, such as food scraps, yard trimmings, paper, wood, leather, etc.
- Organic waste materials can be converted into useful products, such as compost, biogas, biofuels, bioplastics, etc., through various processes, such as composting, anaerobic digestion, fermentation, pyrolysis, etc.
- Composting is the natural decomposition of organic waste materials by microorganisms under aerobic conditions, resulting in a nutrient-rich soil amendment that can improve soil quality and fertility.
- Anaerobic digestion is the breakdown of organic waste materials by microorganisms under anaerobic conditions, producing biogas (a mixture of methane and carbon dioxide) and digestate (a solid or liquid residue that can be used as fertilizer or soil conditioner).
- Fermentation is the conversion of organic waste materials into ethanol or other organic acids by microorganisms under anaerobic or low-oxygen conditions, which can be used as biofuels or chemical feedstocks.
- Pyrolysis is the thermal decomposition of organic waste materials at high temperatures in the absence of oxygen, producing biochar (a carbon-rich solid that can enhance soil properties and sequester carbon), bio-oil (a liquid that can be refined into biofuels or chemicals), and syngas (a gas that can be used for heat or power generation).
- The production of useful products from organic waste materials can have various environmental and social benefits, such as:
 - Reducing the amount of waste that goes to landfills or dumpsites, which can emit greenhouse gases, leachate, and odors, and pose health and safety risks .
 - Saving natural resources and energy that would otherwise be used to produce synthetic fertilizers, fuels, plastics, etc., and reducing the associated environmental impacts, such as greenhouse gas emissions, water pollution, soil degradation, etc. .
 - Enhancing soil health and productivity by adding organic matter and nutrients, improving water retention and infiltration, and suppressing pests and diseases .
 - Providing renewable and clean sources of energy that can reduce dependence on fossil fuels and contribute to climate change mitigation and adaptation .
 - Creating economic and social opportunities for waste management, product development, and value addition, especially for rural and marginalized communities .

Unit-5: 8

- This unit covers the topic of **8-bit microprocessors**, which are electronic devices that can perform arithmetic, logic, and control operations on 8-bit data.
- The unit introduces the basic concepts of microprocessor architecture, instruction set, addressing modes, and interfacing techniques.
- The unit also discusses the features and applications of the **Intel 8085 microprocessor**, which is one of the most popular 8-bit microprocessors in the history of computing.
- The unit consists of the following subtopics:
 - 8.1 Introduction to Microprocessors
 - * Definition and classification of microprocessors
 - * Evolution and generations of microprocessors
 - * Advantages and disadvantages of microprocessors
 - * Applications of microprocessors
 - 8.2 Architecture of 8085 Microprocessor
 - * Block diagram and pin diagram of 8085 microprocessor
 - * Registers and flags of 8085 microprocessor
 - * Bus system and bus signals of 8085 microprocessor
 - * Clock and reset circuits of 8085 microprocessor
 - 8.3 Instruction Set of 8085 Microprocessor
 - * Types and formats of instructions
 - * Opcode and operand of instructions
 - * Classification and examples of instructions
 - * Instruction cycle and timing diagram of instructions
 - 8.4 Addressing Modes of 8085 Microprocessor
 - * Definition and types of addressing modes
 - * Advantages and disadvantages of addressing modes
 - * Examples and calculations of addressing modes
 - 8.5 Interfacing of 8085 Microprocessor
 - * Definition and need of interfacing
 - * Types and examples of interfacing devices
 - * Memory and I/O interfacing techniques
 - * Interrupts and DMA in 8085 microprocessor

Materials Chemistry

- Materials chemistry is a branch of chemistry that deals with the design, synthesis, characterization, processing and understanding of materials with interesting or useful physical properties .
- Materials can be classified into different types based on their composition, structure, properties and applications. Some common types of materials

are metals, ceramics, polymers, composites, biomaterials, nanomaterials, etc.

- Materials chemistry aims to create new forms of matter, whether organic, inorganic or hybrid, that can perform specific functions or have novel characteristics, such as magnetism, optical activity, catalysis, biocompatibility, etc .
- Materials chemistry also seeks to understand the molecular-level mechanisms that govern the behavior and interactions of materials, such as bonding, phase transitions, diffusion, reaction kinetics, etc.
- Materials chemistry is an interdisciplinary field that draws from various branches of chemistry, such as organic, inorganic, physical, analytical and biochemistry, as well as from other disciplines, such as physics, engineering, biology and mathematics.
- Materials chemistry has many applications in various domains, such as energy, electronics, medicine, environment, aerospace, etc. Some examples of materials chemistry products are batteries, solar cells, LEDs, sensors, drug delivery systems, biodegradable plastics, etc.

Polymers; Classification, Polymerization processes, Thermosetting and Thermoplastic

- Polymers are large molecules composed of repeating units called monomers, which are linked by covalent bonds.
- Polymers can be classified based on their origin, structure, molecular forces, and properties.
- Based on their origin, polymers can be natural (such as cellulose, starch, silk, wool, etc.) or synthetic (such as nylon, polyethylene, polystyrene, etc.).
- Based on their structure, polymers can be linear (such as polyethylene, PVC, etc.), branched (such as amylopectin, low-density polyethylene, etc.), or cross-linked (such as vulcanized rubber, bakelite, etc.).
- Based on their molecular forces, polymers can be elastomers (such as rubber, latex, etc.), fibres (such as nylon, polyester, etc.), thermoplastics (such as polyethylene, polystyrene, etc.), or thermosets (such as epoxy, phenolic, etc.).
- Based on their properties, polymers can be homopolymers (made of one type of monomer, such as polyethylene, polystyrene, etc.) or copolymers (made of two or more types of monomers, such as nylon, polyethylene terephthalate, etc.).
- Polymerization is the process of forming polymers from monomers. There are two main types of polymerization: addition and condensation.
- Addition polymerization is the process of forming polymers by adding monomers to a growing chain without losing any atoms or molecules. The monomers must have a double or triple bond that can be broken to form

new bonds with other monomers. For example, ethylene can undergo addition polymerization to form polyethylene.

- Condensation polymerization is the process of forming polymers by combining monomers with the elimination of a small molecule, such as water, ammonia, or hydrogen chloride. The monomers must have two functional groups that can react with each other to form a new bond and release a small molecule. For example, ethylene glycol and terephthalic acid can undergo condensation polymerization to form polyethylene terephthalate and water.
- Thermosetting polymers are polymers that can be cured or hardened by heat, pressure, or chemical agents. They have a highly cross-linked structure that prevents them from melting or flowing when heated. They are rigid, strong, and resistant to heat and chemicals. They cannot be recycled or reshaped. Examples of thermosetting polymers are epoxy, phenolic, and urea-formaldehyde.
- Thermoplastic polymers are polymers that can be softened or melted by heat and reshaped or remolded by cooling. They have a linear or branched structure that allows them to flow when heated. They are flexible, ductile, and recyclable. Examples of thermoplastic polymers are polyethylene, polystyrene, and PVC.

Polymers, Polymer Blends and Composites, Conducting and Biodegradable polymers

- Polymers are large molecules composed of repeating units called monomers, which are linked by covalent bonds. Polymers can have various properties depending on their structure, composition, molecular weight, and processing methods. Some common examples of polymers are plastics, rubber, fibers, and biopolymers.
- Polymer blends are mixtures of two or more polymers that are physically mixed but not chemically bonded. Polymer blends can improve the properties of individual polymers by combining their advantages and minimizing their disadvantages. For example, polymer blends can enhance the mechanical strength, thermal stability, biodegradability, or compatibility of polymers. Some common examples of polymer blends are polyethylene/polypropylene, polystyrene/poly(methyl methacrylate), and poly(lactic acid)/poly(butylene adipate-co-terephthalate).
- Polymer composites are materials that consist of a polymer matrix and a reinforcing filler, such as fibers, particles, or nanomaterials. Polymer composites can have superior properties than the polymer matrix alone, such as high stiffness, strength, toughness, conductivity, or biocompatibility. Some common examples of polymer composites are carbon fiber reinforced plastics, wood plastic composites, and graphene/polymer nanocomposites.
- Conducting polymers are polymers that can conduct electricity due to the

presence of conjugated double bonds or delocalized electrons along the polymer backbone. Conducting polymers can have various applications in electronics, sensors, actuators, batteries, and biomedical devices. Some common examples of conducting polymers are polyaniline, polypyrrole, and polythiophene.

- Biodegradable polymers are polymers that can be degraded by natural processes, such as hydrolysis, enzymatic cleavage, or microbial action. Biodegradable polymers can reduce the environmental impact of plastic waste and can be used for biomedical purposes, such as drug delivery, tissue engineering, or wound healing. Some common examples of biodegradable polymers are poly(lactic acid), poly(glycolic acid), and poly(caprolactone).

Preparation, properties, industrial applications of Teflon, Lucite, Bakelite, Kelvar, Dacron

Teflon

- Teflon is a synthetic polymer of tetrafluoroethylene (C_2F_4) with the formula $(\text{C}_2\text{F}_4)_n$.
- It is prepared by free radical polymerization of tetrafluoroethylene in a pressurized vessel at high temperature with a catalyst such as peroxide or persulfate.
- It has the following properties:
 - It is a white solid compound at room temperature with a density of about 2.2 g/cm^3 .
 - It has a high melting point of about 600 K and a low glass transition temperature of about 130 K .
 - It is chemically inert and resistant to most acids, bases, solvents, and oxidizing agents. The only chemicals that can affect it are alkali metals and some fluorinating agents.
 - It has a low coefficient of friction and a high dielectric strength, making it suitable for non-stick coatings and electrical insulation.
 - It has a low water absorption capacity and a high thermal stability, allowing it to withstand extreme temperatures and humidity.
- It has the following industrial applications:
 - It is widely used as a non-stick coating for cookware, bakeware, and other utensils.
 - It is also used as a lubricant for bearings, gears, valves, and other mechanical parts.
 - It is employed as a protective coating for wires, cables, connectors, and printed circuit boards in the electrical and electronics industry.
 - It is also used as a fabric protector for clothing, carpets, upholstery, and other textiles, as it repels water, oil, and stains.
 - It is also used as a component of some medical devices, such as

catheters, sutures, implants, and artificial joints, as it is biocompatible and resistant to infection.

Lucite

- Lucite is a trade name for polymethyl methacrylate (PMMA), a transparent thermoplastic polymer of methyl methacrylate (MMA) with the formula $(C_5O_2H_8)_n$.
- It is prepared by free radical polymerization of methyl methacrylate in a bulk or suspension process with a catalyst such as benzoyl peroxide or azobisisobutyronitrile.
- It has the following properties:
 - It is a clear, colorless, and lightweight material with a density of about 1.2 g/cm³.
 - It has a high refractive index of about 1.49 and a low dispersion of about 0.009, making it suitable for optical applications.
 - It has a good resistance to ultraviolet light, weathering, and aging, but it is prone to scratching and cracking.
 - It has a moderate mechanical strength and rigidity, but it is brittle and has a low impact resistance.
 - It has a low thermal conductivity and a high thermal expansion coefficient, making it susceptible to thermal stress and deformation.
- It has the following industrial applications:
 - It is widely used as a substitute for glass in windows, skylights, lenses, mirrors, and other transparent objects.
 - It is also used as a material for signs, displays, sculptures, jewelry, and other decorative items.
 - It is also used as a component of some medical devices, such as dentures, artificial eyes, and intraocular lenses, as it is biocompatible and easy to mold.
 - It is also used as a matrix for some composite materials, such as carbon fiber reinforced plastic (CFRP) and glass fiber reinforced plastic (GFRP), as it provides transparency and strength.

Bakelite

- Bakelite is a trade name for phenol-formaldehyde resin, a thermosetting polymer of phenol and formaldehyde with the formula $(C_6H_6O-CH_2O)_n$.
- It is prepared by condensation polymerization of phenol and formaldehyde in an acidic or alkaline medium with a catalyst such as zinc chloride, hydrochloric acid, or ammonia.
- It has the following properties:
 - It is a dark brown or black solid material with a density of about 1.4 g/cm³.
 - It has a high melting point of about 800 K and a high decomposition temperature of about 1000 K, making it suitable for high-

temperature applications.

Thiokol, Nylon, Buna-N and Buna-S and their environmental impact on society, Speciality

- Thiokol is a synthetic rubber made from ethylene dichloride and sodium polysulfide. It is used as a sealant, adhesive, and binder for solid rocket propellants. It has high resistance to heat, cold, and chemicals, but it is also flammable and toxic.
- Nylon is a synthetic polymer made from coal, water, and air. It is used as a fiber for clothing, carpets, parachutes, and other products. It has high strength, elasticity, and durability, but it is also non-biodegradable and requires a lot of energy and water to produce.
- Buna-N is a synthetic rubber made from acrylonitrile and butadiene. It is also known as nitrile rubber or NBR. It is used as a material for hoses, gloves, seals, and gaskets. It has high resistance to oil, fuel, and other chemicals, but it is also non-renewable and can release harmful substances when burned.
- Buna-S is a synthetic rubber made from styrene and butadiene. It is also known as styrene butadiene rubber or SBR. It is used as a material for tires, footwear, and rubber bands. It has good abrasion and weather resistance, but it is also non-renewable and can cause environmental pollution when disposed of.
- The environmental impact of these synthetic polymers on society is complex and depends on various factors, such as their production, use, and disposal methods, their alternatives, and their benefits and drawbacks. Some of the general impacts are:
 - They reduce the dependence on natural resources, such as rubber, cotton, and silk, which may have limited availability, high cost, or ethical issues.
 - They provide new and improved properties, such as strength, durability, flexibility, and resistance, which may enhance the performance, safety, and comfort of various products and applications.
 - They consume a lot of energy, water, and raw materials, which may deplete the natural resources, increase the greenhouse gas emissions, and contribute to climate change.
 - They generate a lot of waste, which may accumulate in landfills, oceans, and soil, and harm the wildlife, ecosystems, and human health.
 - They may release toxic substances, such as dioxins, furans, and heavy metals, during their production, use, or disposal, which may contaminate the air, water, and food, and cause cancer, hormonal disruption,

and other diseases.

- The speciality of these synthetic polymers is that they are man-made materials that can be designed and modified to suit specific needs and purposes. They can also be combined with other materials, such as fillers, additives, and reinforcements, to create composites with enhanced properties and functions.

: <https://en.wikipedia.org/wiki/Thiokol> <https://www.treehugger.com/what-is-nylon-and-is-it-sustainable-5086614>
https://en.wikipedia.org/wiki/Nitrile_rubber
<https://www.icis.com/explore/resources/news/2008/05/12/9122056/history-of-the-synthetic-rubber-industry/>
<https://www.bbc.co.uk/bitesize/guides/zq4m7ty/revision/1>
<https://www.bbc.co.uk/bitesize/guides/zq4m7ty/revision/2>
<https://www.bbc.co.uk/bitesize/guides/zq4m7ty/revision/3>
<https://www.bbc.co.uk/bitesize/guides/zq4m7ty/revision/4>

Polymers

- Polymers are any class of natural or synthetic substances composed of very large molecules, called macromolecules, that are multiples of simpler chemical units called monomers.
- Examples of polymers are rubber, plastics, and nylon.
- Polymers have different physical and chemical properties, which are affected by the structure, type of monomer units, and other factors.
- Some of the properties of polymers are:
 - Density: The mass per unit volume of a polymer. It depends on the molecular weight, degree of crystallinity, and packing of chains.
 - Melting point: The temperature at which a polymer changes from a solid to a liquid state. It depends on the intermolecular forces, degree of crystallinity, and molecular weight of the polymer.
 - Glass transition temperature: The temperature at which a polymer changes from a rigid to a rubbery state. It depends on the flexibility, polarity, and molecular weight of the polymer.
 - Tensile strength: The maximum stress that a polymer can withstand before breaking. It depends on the orientation, cross-linking, and molecular weight of the polymer.
 - Elasticity: The ability of a polymer to return to its original shape after being stretched or compressed. It depends on the cross-linking, molecular weight, and degree of crystallinity of the polymer.
 - Solubility: The ability of a polymer to dissolve in a solvent. It depends on the polarity, molecular weight, and branching of the polymer.
- Polymers can be classified into different types based on their source, structure, and properties :

- Natural polymers: Polymers that are found in nature, such as proteins, cellulose, and nucleic acids.
- Synthetic polymers: Polymers that are man-made, such as plastics, rubber, and nylon.
- Homopolymers: Polymers that are made of only one type of monomer, such as polyethylene and polystyrene.
- Copolymers: Polymers that are made of two or more types of monomers, such as nylon and polyester.
- Linear polymers: Polymers that have long and straight chains of monomers, such as polyethylene and nylon.
- Branched polymers: Polymers that have chains of monomers with side branches, such as polypropylene and amylopectin.
- Cross-linked polymers: Polymers that have chains of monomers that are linked by covalent bonds, such as rubber and bakelite.
- Thermoplastics: Polymers that can be melted and reshaped by heating, such as polyethylene and polystyrene.
- Thermosets: Polymers that cannot be melted and reshaped by heating, as they form irreversible cross-links, such as rubber and bakelite.

Organometallic Compounds: General methods of preparation and applications of

Organometallic compounds are substances that contain one or more metal-carbon bonds, where the carbon is part of an organic group. Organometallic compounds have played a major role in the development of chemistry, as they are used as catalysts, reagents, and intermediates in various reactions.

General methods of preparation

There are four main methods of preparing organometallic compounds, which involve the formation of a metal-carbon bond:

- **Reaction with metal and transmetallation:** This method involves the direct reaction of a metal with an organic halide, or the exchange of a metal with another metal in an organometallic compound. For example, the reaction of magnesium with bromobenzene produces phenylmagnesium bromide, a Grignard reagent, which can then react with copper(I) iodide to form phenylcopper, a Gilman reagent.
- **Metal displacement:** This method involves the displacement of a less electropositive metal by a more electropositive metal in an organic compound. For example, the reaction of sodium with acetylene produces sodium acetylide, which can then react with an alkyl halide to form an alkyne.
- **Metathesis:** This method involves the exchange of ligands between two

organometallic compounds, or between an organometallic compound and a metal salt. For example, the reaction of lithium methyl with cadmium chloride produces methylcadmium and lithium chloride, which can then react with another alkyl halide to form a dialkylcadmium compound.

- **Hydrometallation:** This method involves the addition of a metal hydride to an unsaturated organic compound, such as an alkene or an alkyne. For example, the reaction of lithium aluminium hydride with ethylene produces ethylaluminium dihydride, which can then react with an aldehyde or a ketone to form an alcohol.

Applications of organometallic compounds

Organometallic compounds have various applications in organic synthesis, as they can act as nucleophiles, electrophiles, or catalysts. Some examples of their applications are:

- **Grignard reagents:** These are organomagnesium compounds that can react with various electrophiles, such as aldehydes, ketones, esters, nitriles, and halides, to form alcohols, amines, and alkylated products.
- **Gilman reagents:** These are organocopper compounds that can react with alkyl halides, vinyl halides, and aryl halides, to form carbon-carbon bonds via coupling reactions.
- **Wittig reagents:** These are phosphonium ylides that can react with aldehydes and ketones to form alkenes via the Wittig reaction.
- **Ziegler-Natta catalysts:** These are complexes of transition metals, such as titanium, zirconium, or nickel, with organoaluminium compounds, that can catalyze the polymerization of alkenes to form polyolefins, such as polyethylene and polypropylene.
- **Ferrocene and its derivatives:** These are sandwich compounds that consist of two cyclopentadienyl rings bonded to a central iron atom. They have applications in medicinal chemistry, as they can act as antimalarial, anticancer, and antidiabetic agents, as well as in materials science, as they can act as conductors, sensors, and catalysts.

Organometallic Compounds (RMgX and LiAlH_4)

- Organometallic compounds are compounds that contain a metal-carbon bond, R-M , where R is an organic group and M is a metal.
- Organometallic compounds of magnesium (RMgX) and lithium aluminium hydride (LiAlH_4) are important reagents in organic synthesis, as they can act as nucleophiles and reduce carbonyl groups to alcohols.
- RMgX compounds are also known as Grignard reagents, named after Victor Grignard, who discovered them in 1900 .

- LiAlH_4 is also known as LAH, and was first prepared by Finholt, Bond and Schlesinger in 1947.

Preparation of RMgX and LiAlH_4

- RMgX compounds are prepared by reacting an alkyl or aryl halide (R-X) with magnesium metal (Mg) in an ether solvent, such as diethyl ether or tetrahydrofuran (THF).
- The reaction is highly exothermic and requires an initiation step, such as adding a small amount of iodine (I_2) or a preformed Grignard reagent, to start the reaction.
- The reaction mechanism involves a radical chain process, where the halogen atom (X) is transferred from R-X to Mg , forming $\text{R}^\bullet$ and $\text{MgX}^\bullet$ radicals, which then combine to form RMgX .
- LiAlH_4 is prepared by reacting lithium hydride (LiH) with aluminium chloride (AlCl_3) in an ether solvent, such as diethyl ether or THF.
- The reaction is also highly exothermic and requires cooling and stirring to control the temperature and prevent decomposition of LiAlH_4 .
- The reaction mechanism involves a nucleophilic attack of LiH on AlCl_3 , forming LiAlH_3Cl and LiCl , followed by another attack of LiH on LiAlH_3Cl , forming LiAlH_4 and LiCl .

Structure and Properties of RMgX and LiAlH_4

- RMgX compounds are tetrahedral in solution, as the magnesium atom (Mg) is surrounded by two organic groups (R) and two halide atoms (X).
- The metal-carbon bond in RMgX is polarized, as the carbon atom (C) is more electronegative than the magnesium atom (Mg), making the carbon atom partially negative ($-$) and the magnesium atom partially positive ($+$).
- The carbon atom in RMgX is also sp^3 -hybridized, giving it a tetrahedral geometry and a bond angle of about 109.5° .
- LiAlH_4 is a complex hydride, as it contains four hydride ions (H^-) bonded to a central aluminium atom (Al).
- The structure of LiAlH_4 is tetrahedral, as the aluminium atom (Al) is surrounded by four hydride atoms (H).
- The metal-hydrogen bond in LiAlH_4 is also polarized, as the hydrogen atom (H) is more electronegative than the aluminium atom (Al), making the hydrogen atom partially negative ($-$) and the aluminium atom partially positive ($+$).
- The aluminium atom in LiAlH_4 is also sp^3 -hybridized, giving it a tetrahedral geometry and a bond angle of about 109.5° .

Reactions of RMgX and LiAlH₄

- RMgX compounds are strong nucleophiles and bases, as they can donate a pair of electrons from the carbon atom (C) to an electrophile or a proton (H⁺).
- RMgX compounds can react with various carbonyl compounds, such as aldehydes, ketones, esters, and acid chlorides, to form alcohols on hydrolysis.
- The reaction mechanism involves a nucleophilic addition of RMgX to the electrophilic carbon (C) of the carbonyl group (C=O), forming a tetrahedral intermediate, which then reacts with water (H₂O) to form the alcohol (R-OH) and the magnesium salt (MgX₂).
- RMgX compounds can also react with other electrophiles, such as epoxides, nitriles, and halides, to form carbon-carbon bonds.
- LiAlH₄ is a strong reducing agent, as it can donate a hydride

Course Outcomes:

- Course outcomes are statements that describe what students are expected to know and be able to do by the end of a course.
- Course outcomes are aligned with the course objectives, which are derived from the program outcomes and the institutional mission and vision.
- Course outcomes are measurable, observable, and achievable within the duration and scope of the course.
- Course outcomes are written in terms of student learning, using action verbs that indicate the level of cognitive skills required.
- Course outcomes are communicated to the students at the beginning of the course and throughout the course delivery.
- Course outcomes are assessed using appropriate methods and tools that match the intended learning outcomes and the level of cognitive skills.
- Course outcomes are used to evaluate the effectiveness of the course design, delivery, and assessment, and to identify areas for improvement.

Upon completion of the course the student should be able to:

- Identify and explain the main concepts, theories and methods of sociology.
- Apply sociological perspectives and analytical tools to various social phenomena and issues.
- Compare and contrast different sociological approaches and paradigms.
- Evaluate the strengths and limitations of different sources of sociological data and evidence.
- Conduct basic sociological research using appropriate methods and ethical principles.
- Communicate sociological arguments and findings effectively in written and oral forms.
- Demonstrate critical thinking and reflexivity in relation to their own and others' social positions and experiences.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to discuss. You have asked me to write on the topic of units course outcomes Bloom's. Here is what I have written:

Units Course Outcomes Bloom's

- Units course outcomes are statements that describe what students are expected to know and be able to do by the end of a unit of study.
- Bloom's taxonomy is a framework that classifies learning objectives into six levels of cognitive complexity: remember, understand, apply, analyze, evaluate, and create.
- Units course outcomes should be aligned with Bloom's taxonomy to ensure that students achieve different levels of learning and demonstrate higher-order thinking skills.
- Some examples of units course outcomes and their corresponding Bloom's levels are:
 - By the end of this unit, students will be able to **remember** the definitions and formulas of basic concepts in physics. (Remember)
 - By the end of this unit, students will be able to **understand** the main principles and laws of thermodynamics and how they relate to real-world phenomena. (Understand)
 - By the end of this unit, students will be able to **apply** the concepts and methods of calculus to solve problems involving rates of change, optimization, and integration. (Apply)
 - By the end of this unit, students will be able to **analyze** the structure and function of different types of cells and tissues in the human body. (Analyze)
 - By the end of this unit, students will be able to **evaluate** the strengths and weaknesses of different ethical theories and arguments. (Evaluate)
 - By the end of this unit, students will be able to **create** their own research questions and hypotheses based on the literature review and data collection methods. (Create)
- Units course outcomes should be specific, measurable, achievable, relevant, and time-bound (SMART) to ensure that they are clear, realistic, and assessable.
- Units course outcomes should be communicated to students at the beginning of the unit and reviewed throughout the unit to monitor their progress and provide feedback.

Level

- A level is a device or instrument that measures the inclination or slope of a surface relative to the horizontal or vertical plane.
- A level can be used for various purposes, such as construction, surveying, engineering, carpentry, landscaping, etc.
- A level can also refer to a horizontal or flat surface, such as the ground, a floor, a table, etc.
- There are different types of levels, such as spirit levels, laser levels, digital levels, water levels, etc.
- A spirit level consists of a sealed glass tube filled with a liquid, usually alcohol or water, with an air bubble inside. The tube is mounted on a frame with markings to indicate the center or zero point. When the frame is placed on a surface, the bubble moves to the center if the surface is level, or to one side if the surface is tilted.
- A laser level projects a beam of light that can be used to align objects or mark a straight line on a wall or floor. Some laser levels can also measure angles or distances using sensors or receivers.
- A digital level uses an electronic sensor to detect the angle of inclination and displays it on a screen. Some digital levels can also store data, calculate slopes, or communicate with other devices via Bluetooth or Wi-Fi.
- A water level consists of a flexible tube filled with water, with a transparent container or gauge at each end. The water level indicates the same height at both ends, regardless of the shape or elevation of the tube. A water level can be used to measure the difference in height between two points or to transfer a level from one place to another.

Hello, I am Sydney, your AI assistant. I can help you with your topic. Here is some content in markdown format that you can use to study for your exams.

CO-1 Get an understanding of the theoretical principles of chemistry of molecular structure, bonding and

- Molecular structure refers to the arrangement of atoms in a molecule and the chemical bonds that hold them together.
- Bonding is the process of forming chemical bonds between atoms to create molecules or compounds.
- Theoretical principles of chemistry of molecular structure and bonding include:
 - The octet rule: atoms tend to form bonds that give them eight electrons in their valence shell, or outermost energy level.

- The Lewis structure: a representation of the valence electrons of atoms using dots and lines to show bonds and lone pairs.
- The VSEPR theory: the shape of a molecule is determined by the repulsion between the electron pairs around the central atom.
- The hybridization theory: the orbitals of the central atom mix to form hybrid orbitals that can accommodate the bonding and non-bonding electron pairs.
- The molecular orbital theory: the orbitals of the atoms in a molecule combine to form molecular orbitals that can be occupied by electrons and determine the bond order, bond length, and bond energy.
- The valence bond theory: the bonds between atoms are formed by the overlap of atomic orbitals that contain unpaired electrons.
- The electronegativity: the ability of an atom to attract electrons in a bond. It affects the polarity, dipole moment, and intermolecular forces of a molecule.
- The resonance: the phenomenon of having more than one valid Lewis structure for a molecule that differs only in the distribution of electrons. It affects the stability, bond length, and bond energy of a molecule.

Properties and Chemistry of Advanced Materials (Liquid Crystals, Nanomaterials, Graphite and Fullerene)

- Liquid crystals are substances that exhibit both fluidity and long-range order. They can be classified into thermotropic (temperature-dependent) and lyotropic (concentration-dependent) types. Liquid crystals have applications in displays, sensors, optical devices, and drug delivery systems.
- Nanomaterials are materials that have at least one dimension in the range of 1 to 100 nanometers. They have unique physical, chemical, optical, and electrical properties that depend on their size, shape, synthesis conditions, and functionalization. Nanomaterials have applications in electronics, medicine, energy, environment, and catalysis.
- Graphite is an allotrope of carbon that consists of layers of hexagonal rings of carbon atoms. Graphite has a high electrical conductivity, thermal stability, lubricity, and mechanical strength. Graphite has applications in batteries, electrodes, lubricants, pencils, and graphene production.
- Fullerene is another allotrope of carbon that consists of spherical, cylindrical, or ellipsoidal cages of carbon atoms. Fullerene has a high electron affinity, versatility, and structural diversity. Fullerene has applications in nanotechnology, drug delivery, solar cells, and superconductors.

Principles of Green Chemistry

Green chemistry is the approach in chemical sciences that efficiently uses renewable raw materials, eliminating waste and avoiding the use of toxic and hazardous reagents and solvents in the manufacture and application of chemical products.

There are twelve basic principles of green chemistry that can be grouped into “Reducing Risk” and “Minimizing the Environmental Footprint”. They are:

- **Prevention:** It is better to prevent waste than to treat or clean up waste after it has been created.
- **Atom Economy:** Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.
- **Less Hazardous Chemical Syntheses:** Wherever practicable, synthetic methods should be designed to use and generate substances that possess little or no toxicity to human health and the environment.
- **Designing Safer Chemicals:** Chemical products should be designed to affect their desired function while minimizing their toxicity.
- **Safer Solvents and Auxiliaries:** The use of auxiliary substances (e.g., solvents, separation agents, etc.) should be made unnecessary wherever possible and innocuous when used.
- **Design for Energy Efficiency:** Energy requirements of chemical processes should be recognized for their environmental and economic impacts and should be minimized. If possible, synthetic methods should be conducted at ambient temperature and pressure.
- **Use of Renewable Feedstocks:** A raw material or feedstock should be renewable rather than depleting whenever technically and economically practicable.
- **Reduce Derivatives:** Unnecessary derivatization (use of blocking groups, protection/ deprotection, temporary modification of physical/chemical processes) should be minimized or avoided if possible, because such steps require additional reagents and can generate waste.
- **Catalysis:** Catalytic reagents (as selective as possible) are superior to stoichiometric reagents.
- **Design for Degradation:** Chemical products should be designed so that at the end of their function they break down into innocuous degradation products and do not persist in the environment.
- **Real-time Analysis for Pollution Prevention:** Analytical methodologies need to be further developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances.
- **Inherently Safer Chemistry for Accident Prevention:** Substances and the form of a substance used in a chemical process should be chosen to minimize the potential for chemical accidents, including releases, explosions, and fires.

These principles aim to provide guidance for the design and development of

more sustainable and environmentally friendly chemical processes and products. They also offer opportunities for innovation, cost savings, and social benefits.

K3

- K3 is a type of visa that allows a foreign spouse of a U.S. citizen to enter and reside in the United States while waiting for the approval of their immigrant visa petition.
- K3 visa is a nonimmigrant visa that is valid for two years and can be renewed if the immigrant visa petition is still pending.
- K3 visa holders are eligible to apply for work authorization and travel permission while in the United States.
- K3 visa holders are also eligible to adjust their status to permanent residents once their immigrant visa petition is approved and a visa number is available.
- K3 visa is not available for spouses of lawful permanent residents or for spouses of U.S. citizens who are already in the United States.
- K3 visa requires the U.S. citizen spouse to file two petitions: Form I-130, Petition for Alien Relative, and Form I-129F, Petition for Alien Fiancé(e).
- K3 visa also requires the foreign spouse to apply for a visa at a U.S. consulate abroad and undergo a medical examination and an interview.
- K3 visa processing time varies depending on the workload of the U.S. consulate and the U.S. Citizenship and Immigration Services (USCIS).
- K3 visa may be terminated if the immigrant visa petition is denied, withdrawn, or approved, or if the marriage is terminated by divorce or annulment.

CO-2 Apply the fundamental concepts of determination of structure with various spectral techniques

- Spectroscopy is the study of how electromagnetic radiation, across a spectrum of different wavelengths, interacts with molecules - and how these interactions can be quantified, analyzed, and ultimately interpreted to gain information about molecular structure.
- The most important spectroscopic techniques for structure determination are ultraviolet and visible spectroscopy, infrared spectroscopy, and nuclear magnetic resonance spectroscopy.
- Ultraviolet and visible spectroscopy (UV-Vis) measures the wavelengths of light that are absorbed by molecules in the range of 200-800 nm. The absorption of light corresponds to the excitation of electrons from lower to higher energy levels. The energy and number of the absorbed photons provide information about the electronic structure and conjugation of the molecule.

- Infrared spectroscopy (IR) measures the wavelengths of light that are absorbed by molecules in the range of 2500-25,000 nm. The absorption of light corresponds to the vibration of bonds between atoms. The frequency and intensity of the absorbed photons provide information about the functional groups and the bond strength of the molecule.
- Nuclear magnetic resonance spectroscopy (NMR) measures the interaction of radio waves with the nuclei of atoms in a magnetic field. The absorption of radio waves corresponds to the transition of nuclei from lower to higher energy states. The frequency and intensity of the absorbed photons provide information about the number, type, and environment of the atoms in the molecule.
- Each spectroscopic technique provides different and complementary information about the structure of a molecule. By combining the data from different spectroscopies, the structure of a molecule can be determined with high accuracy and confidence.

Stereochemistry

Stereochemistry is the branch of chemistry that deals with the three-dimensional (3D) structure of molecules and their effect on physical and chemical properties. Stereochemistry is important for understanding the biological activity, synthesis, and analysis of organic compounds.

Types of Stereoisomers

Stereoisomers are molecules that have the same molecular formula and the same connectivity of atoms, but differ in the spatial arrangement of atoms or groups in 3D space. Stereoisomers can be classified into two main types: enantiomers and diastereomers.

Enantiomers

Enantiomers are stereoisomers that are non-superimposable mirror images of each other. Enantiomers have the same physical and chemical properties, except for their interaction with plane-polarized light and other chiral molecules. Enantiomers rotate plane-polarized light in opposite directions, and are said to have opposite optical activities. Enantiomers are also called optical isomers or chiral isomers.

Diastereomers

Diastereomers are stereoisomers that are not mirror images of each other. Diastereomers have different physical and chemical properties, and may have different optical activities. Diastereomers can be further subdivided into several types, such as cis-trans isomers, conformational isomers, and atropisomers.

Cis-trans isomers

Cis-trans isomers are diastereomers that differ in the orientation of atoms or groups around a double bond or a ring . Cis isomers have the same atoms or groups on the same side of the double bond or the ring, while trans isomers have them on opposite sides . Cis-trans isomers are also called geometric isomers or E-Z isomers.

Conformational isomers

Conformational isomers are diastereomers that differ in the rotation of atoms or groups around a single bond . Conformational isomers can interconvert by bond rotation, and are usually in equilibrium with each other. Conformational isomers are also called rotamers or conformers.

Atropisomers

Atropisomers are diastereomers that differ in the rotation of atoms or groups around a bond that is hindered by steric strain or resonance . Atropisomers cannot interconvert by bond rotation, and are usually separated by physical methods. Atropisomers are also called axial chiral isomers or helical isomers.

Stereoisomerism Examples

Some examples of stereochemistry and stereoisomerism are:

- The drug thalidomide, which was used to treat morning sickness in pregnant women, had two enantiomers: one that was effective and one that caused birth defects.
- The amino acids, which are the building blocks of proteins, are chiral molecules that exist as enantiomers: L-amino acids and D-amino acids. Only L-amino acids are found in natural proteins, and D-amino acids are usually toxic or inactive.
- The sugars, which are the main source of energy for living organisms, are also chiral molecules that exist as enantiomers: D-sugars and L-sugars. Only D-sugars are found in natural carbohydrates, and L-sugars are usually rare or synthetic.
- The cis and trans isomers of butenedioic acid (also known as maleic acid and fumaric acid) have different physical and chemical properties, such as melting point, solubility, and acidity.
- The chair and boat conformations of cyclohexane have different steric and energetic interactions, such as torsional strain, angle strain, and van der Waals strain.
- The binaphthyl molecule, which consists of two naphthalene rings connected by a single bond, can exist as two atropisomers: one that is chiral and one that is achiral. The chiral atropisomer can rotate plane-polarized light, while the achiral atropisomer cannot

K4

- K4 is a **symmetric-key algorithm** that is used to encrypt and decrypt data.
- K4 was developed by **David Chaum** and **Jan Camenisch** in 1990 as a variant of the **Blum-Goldwasser cryptosystem**.
- K4 uses a **one-time pad** to generate a random key stream that is XORed with the plaintext to produce the ciphertext.
- K4 also uses a **public-key algorithm** to encrypt and decrypt the one-time pad, which is sent along with the ciphertext.
- K4 is **provably secure** under the assumption that the **decisional Diffie-Hellman problem** is hard and that the **one-time pad** is truly random and secret.
- K4 has some advantages over other symmetric-key algorithms, such as:
 - It is **efficient** and **fast**, since it only requires XOR operations and modular exponentiations.
 - It is **malleable**, meaning that the ciphertext can be modified without affecting the decryption.
 - It is **non-interactive**, meaning that the sender and the receiver do not need to communicate before or during the encryption and decryption process.
 - It is **deniable**, meaning that the sender and the receiver can deny having participated in the communication, since the one-time pad can be chosen arbitrarily.
- K4 has some disadvantages over other symmetric-key algorithms, such as:
 - It is **not authenticated**, meaning that the ciphertext can be tampered with or forged by an adversary without detection.
 - It is **not robust**, meaning that the decryption can fail if the one-time pad is corrupted or lost.
 - It is **not practical**, meaning that the one-time pad requires a large amount of storage and transmission, and that the public-key algorithm introduces additional complexity and overhead.

CO-3 Utilize the theory of construction of electrodes, batteries and fuel cells in redesigning new engineering

- Electrodes are conductors that allow electric current to flow in and out of a cell, such as a battery or a fuel cell.
- Batteries are devices that store chemical energy and convert it to electrical energy when needed. They consist of one or more cells, each with a positive and a negative electrode and an electrolyte that allows ions to move between them.
- Fuel cells are devices that generate electrical energy by converting the

chemical energy of a fuel and an oxidant, such as hydrogen and oxygen. They consist of two electrodes and an electrolyte, similar to batteries, but they do not store energy and need a continuous supply of fuel and oxidant.

- The theory of construction of electrodes, batteries and fuel cells is based on the principles of electrochemistry, which studies the interconversion of chemical and electrical energy by means of redox reactions.
- The performance of electrodes, batteries and fuel cells depends on factors such as the materials, the design, the operating conditions and the efficiency of the reactions.
- The main challenges in designing new engineering applications of electrodes, batteries and fuel cells are to improve their durability, reliability, safety, cost-effectiveness and environmental impact.
- Some examples of new engineering applications of electrodes, batteries and fuel cells are:
 - Flexible and wearable electrodes for biomedical and electronic devices.
 - Lithium-ion batteries with higher energy density and longer lifespan for electric vehicles and portable electronics.
 - Solid oxide fuel cells with lower operating temperature and higher power density for stationary and mobile power generation.
 - Hydrogen fuel cells with improved catalysts and membranes for clean and renewable energy.

Products and Corrosion

Corrosion is the deterioration of a material, usually a metal, due to a chemical or electrochemical reaction with its environment. Corrosion can cause loss of function, aesthetic damage, safety hazards, and economic losses.

Reasons for Corrosion

Some of the common reasons for corrosion are:

- Exposure to oxygen and moisture, which can form rust on iron and steel.
- Exposure to acidic or alkaline substances, which can dissolve or weaken some metals.
- Exposure to salt water or salt spray, which can accelerate corrosion of metals by increasing the conductivity of the electrolyte.
- Exposure to high temperatures, which can increase the rate of chemical reactions and oxidation of metals.
- Exposure to mechanical stress, abrasion, or erosion, which can damage the protective layer of some metals and expose them to corrosive agents.
- Exposure to galvanic corrosion, which occurs when two different metals are in contact and form an electric circuit, causing one metal to corrode faster than the other.

Methods to Control Corrosion

Some of the common methods to control corrosion are:

- Applying a protective coating, such as paint, varnish, oil, or grease, to the metal surface, which can prevent or reduce the contact between the metal and the corrosive environment.
- Applying a sacrificial anode, such as zinc or magnesium, to the metal surface, which can act as a more reactive metal and corrode instead of the protected metal, creating a galvanic cell.
- Applying a cathodic protection, which involves applying an external electric current to the metal surface, which can make the metal act as a cathode and prevent the oxidation reaction.
- Applying a corrosion inhibitor, which is a chemical substance that can reduce the corrosion rate by interfering with the chemical or electrochemical reactions, such as by forming a protective film, neutralizing the corrosive agents, or scavenging the oxygen.
- Applying a corrosion-resistant material, such as stainless steel, aluminum, or titanium, which can resist corrosion by forming a passive layer of oxide or nitride on the surface, which can prevent further oxidation.

Understanding of Chemistry of Engineering Materials (Cement)

Cement is a widely used engineering material that binds aggregates such as sand and gravel to form concrete. Cement is composed of various chemical compounds that undergo hydration reactions when mixed with water, resulting in the hardening and strength development of the cement paste.

Some of the main chemical compounds in cement are:

- **Calcium silicates:** These are the most abundant compounds in cement, accounting for about 70% of the mass. They are responsible for most of the strength and durability of cement. There are two types of calcium silicates: tricalcium silicate (C3S) and dicalcium silicate (C2S). C3S hydrates faster and contributes more to the early strength of cement, while C2S hydrates slower and contributes more to the long-term strength of cement.
- **Calcium aluminates:** These are the second most abundant compounds in cement, accounting for about 10% of the mass. They are responsible for the setting and initial hardening of cement. There are two types of calcium aluminates: tricalcium aluminate (C3A) and tetracalcium aluminoferrite (C4AF). C3A hydrates rapidly and generates heat, which can cause cracking and shrinkage of cement. C4AF hydrates moderately and contributes to the color and resistance to sulfates of cement.
- **Calcium sulfate:** This is a compound that is added to cement as a set

regulator, accounting for about 5% of the mass. It controls the hydration rate of C3A and prevents flash setting of cement. The most common form of calcium sulfate is gypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), which dehydrates partially to form hemihydrate ($\text{CaSO}_4 \cdot 0.5\text{H}_2\text{O}$) or anhydrite (CaSO_4) during the grinding of cement clinker.

- **Other compounds:** These are minor compounds that are present in cement, accounting for less than 5% of the mass. They include magnesium oxide (MgO), sodium oxide (Na_2O), potassium oxide (K_2O), iron oxide (Fe_2O_3), and others. They affect the properties of cement such as color, fineness, soundness, and resistance to alkali-silica reaction.

K3

- K3 is a type of visa that allows a foreign spouse of a U.S. citizen to enter and reside in the United States while waiting for the approval of their immigrant visa petition.
- K3 visa is a nonimmigrant visa that is valid for two years and can be renewed if the immigrant visa petition is still pending.
- K3 visa holders are eligible to apply for work authorization and travel outside the United States with a valid travel document.
- K3 visa holders are also eligible to apply for adjustment of status to become permanent residents once their immigrant visa petition is approved.
- To qualify for a K3 visa, the foreign spouse must meet the following requirements:
 - Be legally married to a U.S. citizen
 - Have a pending Form I-130, Petition for Alien Relative, filed by the U.S. citizen spouse
 - Have a valid passport
 - Have a medical examination and vaccination record
 - Have a police clearance certificate from all countries where they have lived for more than six months since the age of 16
 - Have evidence of financial support from the U.S. citizen spouse or a joint sponsor
 - Have evidence of a bona fide marriage, such as photos, letters, emails, phone records, etc.
 - Have no criminal or immigration violations that would make them inadmissible to the United States
- To apply for a K3 visa, the foreign spouse must follow these steps:
 - Complete Form DS-160, Online Nonimmigrant Visa Application, and pay the visa fee
 - Submit Form I-129F, Petition for Alien Fiancé(e), to the U.S. Citizenship and Immigration Services (USCIS) and receive a receipt notice
 - Wait for the National Visa Center (NVC) to assign a case number and send instructions on how to proceed
 - Schedule and attend a visa interview at the U.S. embassy or consulate

- in their home country
- Bring all the required documents and evidence to the interview
- Receive the visa stamp on their passport and travel to the United States within six months of visa issuance

CO-4 Develop understanding of the sources, impurities and hardness of water, apply the concepts of

- Water is a vital resource for life and various human activities. It is obtained from different sources, such as surface water (rivers, lakes, ponds, etc.), groundwater (wells, springs, etc.), rainwater, seawater, etc.
- Water from natural sources is not pure. It contains various impurities, such as dissolved gases, minerals, salts, organic matter, microorganisms, etc. These impurities may affect the quality, taste, odor, color, and suitability of water for different purposes.
- Hardness of water is a measure of its ability to form lather with soap. It is caused by the presence of dissolved calcium and magnesium salts, such as bicarbonates, sulphates, chlorides, etc. Hard water has some disadvantages, such as forming scales in boilers, pipes, and appliances, reducing the efficiency of detergents and soap, causing skin irritation, etc.
- The concepts of hardness of water include the following:
 - Types of hardness: Temporary hardness and permanent hardness. Temporary hardness is caused by the presence of bicarbonates of calcium and magnesium, which can be removed by boiling or adding lime. Permanent hardness is caused by the presence of sulphates, chlorides, and nitrates of calcium and magnesium, which cannot be removed by boiling or adding lime.
 - Units of hardness: Parts per million (ppm), grains per gallon (gpg), milligrams per liter (mg/L), degrees Clark ($^{\circ}\text{Cl}$), degrees French ($^{\circ}\text{Fr}$), degrees German ($^{\circ}\text{dH}$), etc. One ppm of hardness is equivalent to one milligram of calcium carbonate (CaCO_3) per liter of water. One gpg of hardness is equivalent to 17.1 ppm of hardness. One $^{\circ}\text{Cl}$ of hardness is equivalent to 14.3 ppm of hardness. One $^{\circ}\text{Fr}$ of hardness is equivalent to 10 ppm of hardness. One $^{\circ}\text{dH}$ of hardness is equivalent to 17.8 ppm of hardness.
 - Methods of removing hardness: Ion exchange, reverse osmosis, distillation, precipitation, etc. Ion exchange is a process of exchanging the calcium and magnesium ions in hard water with sodium or hydrogen ions in a resin. Reverse osmosis is a process of forcing hard water through a semipermeable membrane that allows water molecules to pass but blocks the dissolved salts. Distillation is a process of vapor-

izing hard water and condensing the pure water. Precipitation is a process of adding chemicals, such as lime, soda, zeolite, etc., to hard water to form insoluble precipitates of calcium and magnesium salts, which can be filtered out.

Determination of Calorific Values and Analysis of Coal

- Calorific value is a measure of the amount of energy produced from a unit weight of coal when it is combusted in oxygen.
- Calorific value can be determined by different methods, such as thermal analysis, bomb calorimetry, and laser-induced breakdown spectroscopy .
- Thermal analysis is a technique that measures the changes in physical and chemical properties of coal as a function of temperature or time. It can be used to estimate the calorific value of coal based on the heat effects of combustion reactions.
- Bomb calorimetry is a method that involves the complete combustion of a measured sample of coal in a sealed vessel under high pressure and temperature. It can be used to measure the calorific value of coal directly by the amount of heat released in the process.
- Laser-induced breakdown spectroscopy is a method that involves the generation of a plasma by a laser pulse on the surface of coal and the detection of the emitted light spectra. It can be used to measure the calorific value of coal indirectly by the correlation between the spectral intensity and the elemental composition of coal.
- Analysis of coal is the process of determining the physical and chemical properties of coal, such as moisture, ash, volatile matter, fixed carbon, sulfur, and oxygen. It can be used to evaluate the quality and performance of coal for various applications.
- Analysis of coal can be performed by different methods, such as proximate analysis, ultimate analysis, and specific gravity analysis.
- Proximate analysis is a method that measures the moisture, ash, volatile matter, and fixed carbon content of coal. It can be used to estimate the heating value, combustion behavior, and coking potential of coal.
- Ultimate analysis is a method that measures the carbon, hydrogen, nitrogen, sulfur, and oxygen content of coal. It can be used to calculate the empirical formula, air-fuel ratio, and flue gas composition of coal.
- Specific gravity analysis is a method that measures the density of coal relative to water. It can be used to indicate the rank, maturity, and impurities of coal.

K3

- K3 is a type of visa that allows a foreign spouse of a U.S. citizen to enter and reside in the United States while waiting for the approval of their

immigrant visa petition.

- K3 visa is a nonimmigrant visa that is valid for two years and can be renewed if the immigrant visa petition is still pending.
- K3 visa holders are eligible to apply for work authorization and travel outside the United States with a valid travel document.
- K3 visa holders are also eligible to adjust their status to permanent resident once their immigrant visa petition is approved and a visa number is available.
- To qualify for a K3 visa, the foreign spouse must meet the following requirements:
 - Be legally married to a U.S. citizen
 - Have a pending Form I-130, Petition for Alien Relative, filed by the U.S. citizen spouse
 - Have a valid passport and a completed Form DS-160, Online Nonimmigrant Visa Application
 - Have a Form I-129F, Petition for Alien Fiancé(e), filed by the U.S. citizen spouse and approved by the U.S. Citizenship and Immigration Services (USCIS)
 - Have a medical examination and a police clearance certificate from the country of residence
 - Have evidence of a bona fide marriage and financial support from the U.S. citizen spouse
 - Pay the required fees and attend a visa interview at a U.S. embassy or consulate
- The advantages of a K3 visa are:
 - It allows the foreign spouse to join the U.S. citizen spouse in the United States sooner than waiting for the immigrant visa process
 - It allows the foreign spouse to work and travel legally in the United States
 - It allows the foreign spouse to apply for a green card without leaving the United States
- The disadvantages of a K3 visa are:
 - It is subject to annual numerical limits and may not be available at all times
 - It does not guarantee the approval of the immigrant visa petition or the adjustment of status application
 - It may be terminated if the immigrant visa petition is denied, withdrawn, or revoked
 - It may be invalidated if the marriage ends by divorce, annulment, or death

CO-5 Develop the understanding of Chemical structure of polymers and its effect on their various properties

- Polymers are very long carbon-based molecules that have a wide range of beneficial structural (and sometimes thermal) properties.
- The chemical structure of polymers refers to the arrangement of atoms and bonds in the polymer chains, as well as the degree of branching, cross-linking, and crystallinity.
- The chemical structure of polymers affects their physical properties, such as density, melting point, strength, elasticity, solubility, and reactivity.
- Some of the factors that influence the physical properties of polymers are:
 - Chain length: Longer polymer chains have more intermolecular forces, such as van der Waals forces and hydrogen bonding, which make them stronger and stiffer .
 - Side groups: The presence of different functional groups or atoms attached to the polymer chains can affect the polarity, flexibility, and reactivity of the polymers. For example, polyvinyl chloride (PVC) has chlorine atoms as side groups, which make it more polar and less flexible than polyethylene.
 - Branching: The degree of branching refers to the number of side chains that branch off from the main polymer chain. More branching reduces the ability of the polymer chains to pack closely together, which lowers the density, melting point, and crystallinity of the polymers .
 - Cross-linking: The degree of cross-linking refers to the number of covalent bonds that link different polymer chains together. More cross-linking increases the rigidity, strength, and thermal stability of the polymers, but also reduces their solubility and elasticity. For example, vulcanized rubber is more cross-linked than natural rubber, which makes it more durable and resistant to heat.
 - Crystallinity: The degree of crystallinity refers to the proportion of polymer chains that are arranged in a regular and ordered manner. More crystallinity increases the density, melting point, and strength of the polymers, but also reduces their flexibility and transparency. For example, high-density polyethylene (HDPE) is more crystalline than low-density polyethylene (LDPE), which makes it harder and stronger.
- To change the properties of polymers, different methods can be used, such as:
 - Copolymerization: This is the process of combining two or more different monomers to form a polymer with a mixed chemical structure.

- For example, nylon is a copolymer of adipic acid and hexamethylene-diamine, which gives it both hydrophilic and hydrophobic properties.
- Blending: This is the process of mixing two or more different polymers to form a composite material with improved properties. For example, polycarbonate is blended with acrylonitrile butadiene styrene (ABS) to increase its impact resistance and toughness.
 - Additives: These are substances that are added to polymers to modify their properties, such as plasticizers, stabilizers, fillers, colorants, and flame retardants. For example, plasticizers are added to PVC to make it more flexible and soft.

When used as engineering materials. Understanding the applications of specific polymers and Chemistry

- Polymers are large molecules composed of repeating units called monomers, which are linked together by covalent bonds.
- Polymers can be classified into natural and synthetic polymers, depending on their origin and composition.
- Natural polymers are derived from biological sources, such as proteins, cellulose, starch, and rubber. Synthetic polymers are artificially created by chemical reactions, such as plastics, nylon, and silicone.
- Polymers have diverse physical and chemical properties, such as hardness, flexibility, solubility, biodegradability, and thermal stability, which depend on their molecular structure, size, and arrangement.
- Polymers have many applications in engineering and technology, such as:
 - Aircraft, aerospace, and sports equipment: Polymers are used to make lightweight, strong, and durable materials, such as carbon fiber, epoxy resin, and polyurethane foam.
 - 3D printing plastics: Polymers are used to create complex shapes and structures by depositing layers of molten or solidified material, such as polylactic acid, acrylonitrile butadiene styrene, and polyethylene terephthalate.
 - Biopolymers in molecular recognition: Polymers are used to design molecules that can bind to specific targets, such as enzymes, antibodies, and DNA, for applications in biotechnology, medicine, and nanotechnology.
 - Polymers in holography: Polymers are used to create holograms, which are three-dimensional images formed by the interference of light waves, for applications in security, entertainment, and data storage.
 - Organic polymers used in water purification: Polymers are used to remove contaminants from water, such as heavy metals, organic pollutants, and microorganisms, by adsorption, filtration, or coagulation.

- Printed circuit board substrates: Polymers are used to make the base material for electronic circuits, such as polyimide, epoxy, and polytetrafluoroethylene, which provide electrical insulation, mechanical support, and thermal stability.
- Green chemicals: Polymers and biopolymers: Polymers are used to make environmentally friendly chemicals, such as biodegradable plastics, biofuels, and biocatalysts, which reduce the use of fossil fuels, toxic substances, and waste generation.
- Polymeric biomolecules: Polymers are used to study and manipulate biological molecules, such as DNA, RNA, and proteins, which are essential for life and perform various functions in cells.

Industrial Process

An industrial process is a series of operations that transform raw materials or inputs into finished products or outputs. Industrial processes can be classified into different categories, such as:

- Chemical processes: These involve chemical reactions or changes in the molecular structure of the inputs, such as refining, polymerization, fermentation, etc.
- Physical processes: These involve changes in the physical properties or states of the inputs, such as melting, freezing, evaporation, distillation, etc.
- Biological processes: These involve the use of living organisms or enzymes to produce or modify the inputs, such as brewing, cheese making, bioremediation, etc.
- Mechanical processes: These involve the use of machines or tools to shape, cut, join, or separate the inputs, such as machining, welding, forging, etc.
- Electrical processes: These involve the use of electricity or electromagnetic fields to generate, transmit, or manipulate the inputs, such as electroplating, induction heating, laser cutting, etc.

Some examples of industrial processes are:

- Steel making: This is a chemical and physical process that converts iron ore, coke, and limestone into steel, which is an alloy of iron and carbon. The main steps are: 1) Blast furnace: The iron ore, coke, and limestone are heated and reduced to produce molten iron. 2) Basic oxygen furnace: The molten iron is mixed with oxygen and scrap steel to remove impurities and adjust the carbon content. 3) Continuous casting: The molten steel is solidified into slabs, billets, or blooms, which are then rolled or forged into various shapes and sizes.
- Paper making: This is a mechanical and chemical process that converts wood, pulp, or recycled paper into paper or paperboard. The main steps are: 1) Pulping: The wood, pulp, or recycled paper is shredded and mixed

with water and chemicals to separate the fibers. 2) Screening and cleaning: The pulp is filtered and washed to remove contaminants and unwanted materials. 3) Bleaching: The pulp is treated with chemicals to whiten and brighten the fibers. 4) Forming: The pulp is diluted with water and spread over a moving wire mesh to form a thin layer of wet paper. 5) Pressing: The wet paper is squeezed between rollers to remove excess water and improve the strength and smoothness of the paper. 6) Drying: The paper is heated and dried by hot air or steam. 7) Finishing: The paper is coated, calendered, cut, or wound into rolls or sheets, depending on the desired quality and application.

- Cement making: This is a chemical and physical process that converts limestone, clay, and other raw materials into cement, which is a binder for concrete and mortar. The main steps are: 1) Quarrying: The limestone, clay, and other raw materials are mined and crushed. 2) Blending: The raw materials are mixed and ground to form a fine powder called raw meal. 3) Preheating and calcining: The raw meal is heated and partially decomposed in a preheater and a rotary kiln to produce clinker, which is a mixture of calcium silicates and aluminates. 4) Cooling and grinding: The clinker is cooled and ground with gypsum and other additives to produce cement. 5) Packing and shipping: The cement is stored in silos and packed in bags or bulk containers for transportation.

K3

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- K3 visa is a nonimmigrant visa that is valid for two years and can be renewed if the immigrant visa petition is still pending.
- K3 visa holders are eligible to apply for work authorization and travel outside the United States with a valid travel document.
- K3 visa holders are also eligible to adjust their status to permanent resident once their immigrant visa petition is approved and a visa number is available.
- To apply for a K3 visa, the U.S. citizen spouse must file Form I-130, Petition for Alien Relative, and Form I-129F, Petition for Alien Fiancé(e), with the U.S. Citizenship and Immigration Services (USCIS).
- The foreign spouse must then apply for the K3 visa at the U.S. embassy or consulate in their home country, after receiving a notice of action from the USCIS.
- The foreign spouse must also undergo a medical examination and an interview at the U.S. embassy or consulate, and provide the required documents, such as passport, marriage certificate, birth certificate, police clearance, and evidence of relationship with the U.S. citizen spouse.
- The K3 visa processing time varies depending on the USCIS and the U.S.

embassy or consulate, but it usually takes several months to a year or more.

Reference Books:

- Reference books are books that contain factual information on various topics, such as dictionaries, encyclopedias, atlases, directories, handbooks, manuals, etc.
- Reference books are usually consulted for specific information, rather than read cover to cover.
- Reference books are often arranged alphabetically, chronologically, geographically, or by subject, depending on the type and purpose of the book.
- Reference books are usually kept in a separate section of a library, and are not meant to be borrowed or taken out of the library.
- Reference books are useful for finding definitions, facts, statistics, background information, illustrations, maps, charts, etc. on a topic.
- Reference books are updated regularly to reflect the latest information and developments in various fields of knowledge.
- Reference books are often cited as sources of information in academic papers, reports, articles, etc. using a specific citation style, such as APA, MLA, Chicago, etc.

Engineering Chemistry by Rath & Singh, 2nd Edition, Cengage Learning India Pvt Ltd Delhi

- Engineering Chemistry is a textbook that covers the fundamental concepts and applications of chemistry in the field of engineering.
- The book is divided into 12 chapters, each focusing on a specific topic related to engineering chemistry, such as water, fuels, corrosion, polymers, nanomaterials, etc.
- The book aims to provide a comprehensive and updated knowledge of the subject, as well as to develop the analytical and problem-solving skills of the students.
- The book is written in a simple and lucid language, with numerous examples, diagrams, tables, and figures to illustrate the concepts and principles.
- The book also includes multiple-choice questions, review questions, numerical problems, and case studies at the end of each chapter to test the understanding and application of the students.
- The book is suitable for the first-year undergraduate students of engineering and technology, as well as for the students preparing for various competitive examinations.

Engineering Chemistry by SS Dara, S Chand & Co Ltd

- Engineering Chemistry by SS Dara, S Chand & Co Ltd is a textbook for engineering students that covers the basic concepts and applications of chemistry in various fields of engineering.
- The book is divided into 16 chapters, each dealing with a specific topic such as water, fuels, corrosion, polymers, lubricants, electrochemistry, nanotechnology, etc.
- The book provides a comprehensive and systematic presentation of the theoretical and practical aspects of engineering chemistry, with examples, diagrams, tables, and exercises to enhance the understanding of the subject.
- The book also includes multiple choice questions, review questions, and numerical problems at the end of each chapter to test the knowledge and skills of the students.
- The book is suitable for the first year engineering students of various branches, as well as for the students preparing for competitive examinations such as GATE, UPSC, etc.

Engineering Chemistry by Jain & Jain, S.Chand & Comp, New Delhi

- Engineering Chemistry is a textbook written by Jain & Jain and published by S.Chand & Comp, New Delhi.
- The book covers the basic concepts and applications of chemistry in various fields of engineering, such as materials, energy, environment, and biotechnology.
- The book is divided into 16 chapters, each with a summary, review questions, and numerical problems.
- The chapters are as follows:
 1. Atomic Structure and Chemical Bonding
 2. Periodic Properties and Hydrogen
 3. s-Block Elements
 4. p-Block Elements
 5. d- and f-Block Elements
 6. Coordination Compounds and Organometallics
 7. Chemical Thermodynamics
 8. Chemical Kinetics and Catalysis
 9. Electrochemistry
 10. Solutions and Colligative Properties
 11. Phase Equilibria and Phase Rule

12. Chemical Equilibria and Ionic Equilibria
 13. Water and its Treatment
 14. Fuels and Combustion
 15. Polymers and Composites
 16. Corrosion and its Prevention
- The book is suitable for undergraduate students of engineering and technology, as well as for those preparing for competitive examinations.

Engineering Chemistry by K. Sesha Maheswaramma, Pearson

- Engineering Chemistry is a textbook that covers the fundamental concepts and applications of chemistry in the field of engineering.
- The book is divided into four units: Water Technology, Fuels and Combustion, Polymers and Composites, and Corrosion and Surface Engineering.
- The book aims to provide a comprehensive and systematic treatment of the topics, with an emphasis on problem-solving and numerical methods.
- The book also includes numerous examples, exercises, and case studies to illustrate the relevance and practical aspects of chemistry in engineering.
- The book is suitable for undergraduate students of engineering, as well as for professionals and researchers who want to refresh their knowledge of chemistry and its applications.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have chosen the topic of Engineering Chemistry by OG Palanna, Mc Graw Hill Education, New Delhi. Here is a summary of the main points of this book:

- Engineering Chemistry is a textbook that covers the basic concepts and applications of chemistry in various fields of engineering, such as materials, energy, environment, and biotechnology.
- The book is divided into 16 chapters, each focusing on a specific topic, such as atomic structure, chemical bonding, thermodynamics, electrochemistry, corrosion, polymers, fuels, water treatment, nanotechnology, and green chemistry.
- The book provides a clear and concise explanation of the theoretical principles, along with numerous examples, solved problems, and exercises to enhance the understanding and problem-solving skills of the students.
- The book also includes several features, such as learning objectives, key points, summary, review questions, multiple choice questions, and references at the end of each chapter, to help the students revise and assess their learning outcomes.
- The book is designed to meet the syllabus requirements of various engineering courses and examinations, and to cater to the needs of both beginners and advanced learners of chemistry.

Engineering Chemistry by Shashi Chawla

- Engineering Chemistry by Shashi Chawla is a textbook for engineering students of various branches, covering the syllabus of engineering chemistry courses offered by Indian universities .
- The book is divided into 26 chapters, covering topics such as atomic structure, chemical bonding, phase rule, solutions, electrochemistry, corrosion, polymers, fuels, water, instrumental methods of analysis, and environmental chemistry .
- The book aims to provide a clear and concise explanation of the fundamental concepts and principles of chemistry, with an emphasis on their applications in engineering problems.
- The book also contains numerous solved examples, illustrations, tables, and diagrams to aid the understanding of the topics .
- The book is widely recommended in most engineering colleges of India, and has received positive reviews from students and teachers for its comprehensive coverage, simple language, and logical presentation .

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have chosen the topic of University Chemistry by BH Mahan. Here is a summary of the main points of this book:

- University Chemistry by BH Mahan is a textbook that covers the fundamentals of chemistry, such as atomic structure, chemical bonding, thermodynamics, kinetics, equilibrium, electrochemistry, and organic chemistry.
- The book is divided into four parts: Part I: The Structure of Matter, Part II: The Properties of Matter, Part III: Chemical Reactions, and Part IV: Organic Chemistry.
- Part I: The Structure of Matter discusses the basic concepts of atomic and molecular structure, such as the periodic table, quantum theory, orbital hybridization, molecular orbital theory, and valence bond theory. It also introduces the concepts of nuclear chemistry, such as radioactivity, nuclear reactions, and nuclear energy.
- Part II: The Properties of Matter covers the physical and chemical properties of matter, such as the states of matter, intermolecular forces, solutions, colligative properties, phase diagrams, and chemical kinetics. It also explains the principles of thermodynamics, such as the laws of thermodynamics, enthalpy, entropy, free energy, and spontaneity of reactions.
- Part III: Chemical Reactions deals with the types and mechanisms of chemical reactions, such as acid-base reactions, oxidation-reduction reactions, coordination chemistry, and electrochemistry. It also explores the concepts of chemical equilibrium, such as the equilibrium constant, Le Chatelier's principle, solubility product, and common ion effect.
- Part IV: Organic Chemistry introduces the structure, nomenclature, and

reactions of organic compounds, such as alkanes, alkenes, alkynes, aromatic compounds, alcohols, ethers, aldehydes, ketones, carboxylic acids, and amines. It also covers the topics of stereochemistry, such as chirality, optical activity, and conformational analysis. It also discusses the major classes of organic reactions, such as substitution, elimination, addition, and rearrangement reactions.

- University Chemistry by BH Mahan is a comprehensive and rigorous textbook that provides a solid foundation for students who want to pursue higher studies in chemistry or related fields. It is also a useful reference for teachers and researchers who want to refresh their knowledge of chemistry.

University Chemistry by CNR Rao

- University Chemistry by CNR Rao is a textbook for undergraduate students of chemistry, covering the basic concepts and principles of physical, inorganic and organic chemistry.
- The book is divided into three parts: Part I deals with the structure and bonding of atoms and molecules, thermodynamics, chemical kinetics, electrochemistry and surface chemistry. Part II covers the periodic table, coordination chemistry, solid state chemistry, nuclear chemistry and bioinorganic chemistry. Part III introduces the fundamentals of organic chemistry, including hydrocarbons, functional groups, stereochemistry, organic reactions and mechanisms, and natural products.
- The book aims to provide a comprehensive and rigorous treatment of the subject, with an emphasis on problem-solving and numerical methods. It also includes numerous examples, exercises, figures, tables and references to help the students learn and apply the concepts.
- The book is written by CNR Rao, a renowned Indian chemist and professor, who has made significant contributions to the fields of solid state and materials chemistry, nanotechnology, molecular spectroscopy and environmental science. He has authored over 1600 research papers and 50 books, and has received many national and international awards and honors, including the Bharat Ratna, the highest civilian award of India.